

A Map of Tax Responsiveness

Preliminary and incomplete: please do not circulate

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Abstract

Although the applications for which economists need tax elasticities are typically comparative, the methods used to estimate them rest on thin, local, *between-income* contrasts: taxpayers just above and below a threshold, or a tax reform-affected group against a nearby unaffected ones. The variation examined in this paper comes, instead, from differential, *within-income* shifts to the tax liability function. This makes it possible to control for income without absorbing identifying variation, strengthens monotonicity and exogeneity — but also enables, otherwise unfeasible, heterogeneity analyses across the income distribution, spousal income split and gender. Applying the design to the universe of French income tax returns from 2003 to 2023, we find elasticities that decline with income — most steeply at low incomes — a gradient driven by low earners insured by higher-earning partners rather than by the most marginal. The answers to normative questions — whom to tax more, whom less — and to positive ones — what are the effects of tax reforms — are sensitive to this heterogeneity.

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1 Introduction

When a problem is complicated, it can help to zoom in. In the case of tax elasticities, the problem is simultaneity: tax rates are a function of income, and income a function of tax rates, which means that a naive regression of income growth on tax-rate changes does not give causal elasticity estimates. In response, the literature has gone local: (i) zooming in on a kink in the tax schedule, inferring tax elasticities from the excess density of taxpayers just below thresholds—the idea behind bunching designs (H. J. Kleven, 2016; Saez, 2010); or (ii) zooming in on a group affected by a tax reform, approximating its counterfactual earnings growth with a similar, group left unaffected—the idea behind simulated tax-reform IV designs (Gruber & Saez, 2002).

The questions for which elasticities are needed, however, often take the zoom out. Tax elasticity of labour supply (or, “sufficiently”, of taxable income) are key input to optimal-tax models, macro labour supply models, and microsimulations of the effects of reforms - and it may matter which local estimate one draws : an elasticity identified from a cut to top rates need not be close to any representative, relevant average one. And many questions are comparative: the costs of a progressive tax schedule depends on the gradient of the importance of responsiveness across the income distribution, and the costs of a joint tax schedule (taxing married couples income as one), on the ratio of spousal elasticities. The variance of existing estimates is large (Neisser, 2021), and it is difficult to establish how much of it reflects actual heterogeneity in tax responsiveness, and how much the populations, margins, and designs that differ across studies.

This paper proposes a source of variation suited to identifying heterogeneities across income, spousal income split, and gender. Like bunching, the approach exploits discontinuities in the tax schedule, but it does so in the context of “events” that change the tax liability function — like, under family taxation, the fiscal exit of a child from the family tax unit, or marriage. It relies on the fact that, under the French system, differences in partner income make otherwise identical individuals differentially likely to switch tax bracket: two women with the same earnings, e.g. both in the 14% bracket, whose first-born reaches the age at which fiscal unit attachment ends, may see different predicted tax changes — one remaining at 14%, the other, with a bit more high income partner, pushed into the 30% bracket. The idea is to use the

assumption that there is no anticipatory bunching around not yet effective (childless or joint) thresholds.

Two features of this variation make it interesting. First, it comes from outward shifts of otherwise unchanged tax schedules, which makes the monotonicity assumption more plausible than under reform variation. Second, it is within- rather than between-income. When exposure to tax variation is a function of own income — as with a reform affecting a particular part of the income distribution — income cannot be controlled for without absorbing identifying variation, leaving between-income designs exposed to bias from mean reversion and secular trends in income. These are arguably less of a concern when the source of variation is not in own income but driven by constant partner income differences. And because it is present throughout the own- and partner income distribution, some otherwise difficult heterogeneity analyses become feasible, in one same observational setting.

We implement the design using the universe of French tax returns from 2003 to 2023. With newly available individual identifiers, we built an individual-level panel from the tax-unit filings.¹

We find elasticities that decline with income — most steeply at low incomes — and are shifted upward for women; the gradient is driven by low earners insured by higher-earning partners rather than by the most marginal households.

We show the estimated heterogeneity to alter the conclusions of normative models of optimal taxation and of reform assessments. Feeding the estimated income profile into the sufficient-statistics formula for optimal taxation (Saez, 2001), in the implementation of Brewer et al. (2010), makes the optimal schedule substantially steeper than its constant-elasticity benchmark; replacing the constants in the evaluation of reforms towards more individual taxation of couples (Bierbrauer et al., 2024) raises the assessed welfare gains (Section 6).

The paper thereby joins a long-standing effort to understand the dimensions of heterogeneity in tax responsiveness. One dimension is the margin of adjustment — taxable income, broader income concepts, labour supply — where the literature can be summarised as finding that the more comprehensive the income definition considered, the larger the elasticity, with work-income-only elasticities for men barely distinguishable from zero. Another dimension is the

¹Relevant to follow individuals through events such as marriage.

income distribution: studies of multiple reforms (Gruber & Saez, 2002; Jakobsen & Søgaaard, 2022; H. J. Kleven & Schultz, 2014) or of reforms spanning wide income ranges (Graber et al., 2026) generally find larger broad-income responses at the top than in the middle of the distribution. Papers on heterogeneity across gender find married women and single mothers to be above-average elastic in labour supply (Blomquist & Selin, 2010; Eissa, 1995, 1996), and a further strand studies heterogeneity over time, arguing that the gap between macro and micro estimates hinges on different horizons considered (H. Kleven et al., 2025).

The paper also contributes to the methodological effort to clarify the conditions under which, and precisely what, simulated-instrument strategies identify. Gruber and Saez (2002), Weber (2014), Blomquist and Selin (2010), H. J. Kleven and Schultz (2014), and Miao et al. (2024) propose ways to mitigate biases from mean reversion and inequality trends, whose respective merits are discussed by Jakobsen and Søgaaard (2022) and Saez et al. (2012); Kumar and Liang (2020) and Graber et al. (2026) relax the constant-elasticity assumption, and Gruber and Saez (2002) and Graber et al. (2026) separate income from substitution effects. We draw most on Jakobsen and Søgaaard (2022), whose strategies for validating the identifying assumptions we adapt.

The rest of the paper is structured as follows. Section 2 describes the French tax system and the construction of the individual-level panel. Section 3 sets out the theoretical framework, the identification problem, and the event-based instruments we propose. Section 4 develops the empirical strategy and illustrates it with a transparent case study, Section 5 presents the results, and Section 7 concludes.

2 Background

2.1 Institutional Setting

The variation explored in this paper stems from two features of the French tax system: (i) the dependence of the tax liability on household composition through the *family quotient*, and (ii) the step-wise shape of the marginal tax schedule. Together, they imply that a change in household composition can move a household across a bracket threshold—and whether it does depends on the incomes of both spouses.

What makes the French tax system one of “family taxation” is that the tax burden (equation (1)) is a function of the family quotient, which increases with the number of family members: the quotient is 1 for singles, 2 for childless married (or civil-union) couples and 3 for a married couple with two children, increasing for each child up to a cap.²³

$$(1) \quad \text{Tax burden} = T\left(\frac{\text{Taxable income}}{\text{Family quotient}}\right) \times \text{Family quotient}$$

where $T(x)$ denotes the French income tax formula for a single individual.

The resulting marginal tax rates for different family compositions are depicted in Figure 2 Panel A. Panel B illustrates the resulting income tax liabilities, accounting for both the family quotient and the cap on child-related tax advantages.

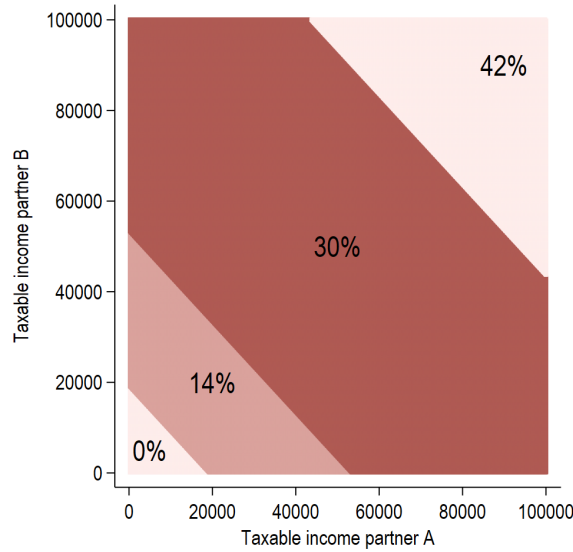
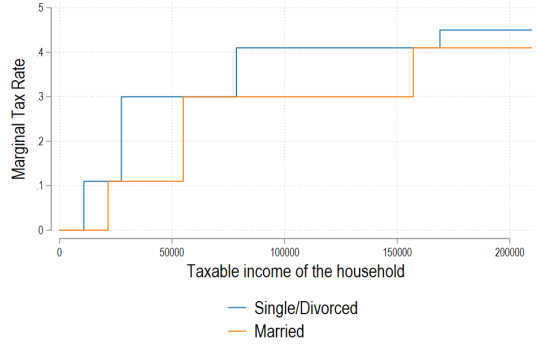


Figure 1: Marginal tax rate as a function of the income of both spouses

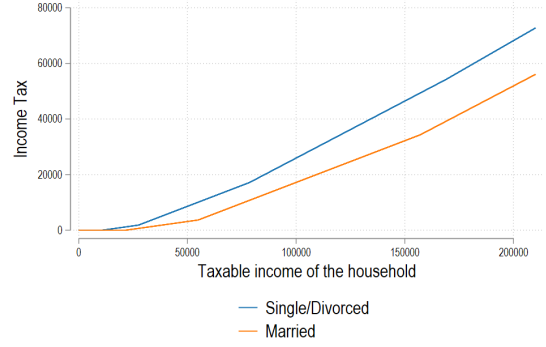
Notes: The heatmap illustrates how the marginal tax rate depends on the income of both partners. Conditional on the income of one partner, the marginal tax rate increases with the income of the other partner. The simulation uses the French tax system of 2015.

²The tax advantage associated with children is capped, such that the benefit from an additional child does not exceed a predetermined threshold.

³Other factors also influence the family quotient. For example, it increases for single parents and parents of disabled children. We do not consider these cases in our analysis.



(a) Marginal tax schedule



(b) Tax burden

Figure 2: Marginal tax schedule and tax burden by household composition

Notes: Panel (a) displays marginal tax rates for different household compositions in 2015. Panel (b) displays the tax burden for different family situations in 2015. Source: Own calculations based on OpenFisca France.

Under French family taxation, the tax function is defined at the household level. In the notation of equation (1), a married or civil-union household with family quotient Q_{it} owes

$$T_t^H(z_{it} + z_{pt}; Q_{it}) = Q_{it} T_t\left(\frac{z_{it} + z_{pt}}{Q_{it}}\right),$$

where z_{pt} is the partner's income. Two features drive everything that follows. First, the marginal rate is common to both spouses:

$$\frac{\partial T_t^H}{\partial z_{it}} = \frac{\partial T_t^H}{\partial z_{pt}} = T_t'\left(\frac{z_{it} + z_{pt}}{Q_{it}}\right), \quad \tau_{it}^H \equiv 1 - T_t'\left(\frac{z_{it} + z_{pt}}{Q_{it}}\right),$$

so that i 's marginal net-of-tax rate depends on partner income—as illustrated in Figure 1—and is an equilibrium object, jointly determined with both spouses' income choices. Second, events that change the household's fiscal configuration shift this function at every level of own income: the fiscal exit of a child lowers Q (from 2.5 to 2 for a one-child couple, Figure 2), and marriage pools the tax base. These are the shocks our design exploits.

2.2 Data

We use the French administrative tax panel known as POTE, provided through secure access by the CASD (Centre d'accès sécurisé aux données).⁴ The dataset covers all income tax returns filed in France between 2003 and 2023.

POTE includes all variables submitted in the tax return (Form No. 2042), intermediate calculations for determining the final tax liability, and all information shown to the household on its official tax notice.⁵

All residents in France are required to file an annual tax return reporting income and family-related information. This results in a total sample of approximately 60 to 68 million individuals per year. Married couples and individuals in civil unions (PACS) must file jointly, except in the year of marriage, when they may opt to file separately.

The dataset contains rich tax-relevant information, including—at the individual level—age, labour income, deductible expenses and calculated tax liabilities. It also includes detailed household-level characteristics such as the number of dependent children, marital status and filing status.

Although the returns are filed at the tax unit level (the *foyer fiscal*), the POTE database has the important advantage of including individual-level identifiers—a recent data innovation. This feature allows the construction of a longitudinal individual-level panel, essential to our design: it lets us link individuals across tax units and follow their incomes before and after family events that change the tax burden, such as marriage or a child's fiscal exit.

The dataset also contains a gender variable, though not available for all tax years.⁶ Other papers use the ordering of the declarant in the form as an indicator of gender, given the first position was historically reserved for “Monsieur”. We use the same approach for the individuals we don't observe in any of the years in which the gender variable is available.

⁴POTE – Panel On The Taxation of Households, INSEE-DGFIP. Data accessed via CASD (<https://www.casd.eu/>).

⁵From 2016 onward, this includes the marginal tax rate explicitly communicated to the household.

⁶Clara to do

3 Estimating tax elasticities

In this section, we (i) set out the theoretical framework and define the estimand, (ii) introduce simulated tax instruments—the reform-based version used in the literature and the event-based version we propose—within that framework, and (iii) extend the framework to a two-earner household to make precise what our variation identifies for individuals in couples. Section 4 then develops the empirical strategy and illustrates it with a transparent case study.

3.1 Theoretical framework

We follow Jakobsen and Søgaaard (2022) and specify a quasi-linear utility function for individual i in year t :

$$u_{it}(c_{it}, z_{it}) = c_{it} - \frac{n_{it}}{1 + \frac{1}{\epsilon}} \left(\frac{z_{it}}{n_{it}} \right)^{1 + \frac{1}{\epsilon}},$$

where c_{it} is consumption and z_{it} is (work / taxable) income; the second term captures the disutility of earning income, governed by the elasticity parameter ϵ and potential income n_{it} .

The budget constraint is

$$c_{it} \leq z_{it} - T_t(z_{it}; Q_{it}),$$

where the tax schedule T_t may depend on household composition through the family quotient Q_{it} of equation (1); for a single filer, $Q_{it} = 1$ and T_t is the individual schedule. Potential income may vary with family events: we write $n_{it} = n_{it}(g_{it})$, where $g_{it} \in \{0, 1\}$ indicates whether the event under study—in our design, the fiscal exit of the first-born child; marriage in the robustness design—has occurred by year t , and leave this dependence unrestricted.⁷ Changes in consumption needs associated with the event enter the budget constraint as a lump-sum term and, with utility linear in consumption, leave the earnings choice unaffected. We record this property as an assumption, since it is what the event-based design must defend:

Assumption S1 (No income effects). *Utility is quasi-linear in consumption: earnings respond to the marginal net-of-tax rate only, and lump-sum changes in tax liability or consumption needs leave the earnings choice unaffected.*

⁷The occurrence and timing of the event are not modelled as choices: the event year m_i —the calendar year in which the first-born turns 21—is fixed by fertility two decades prior. The remaining institutional choice margins (continued fiscal attachment of students, the timing of detachment) are discussed alongside the density tests of Appendix B. For tax reforms, no dependence of n_{it} on the policy change needs to be specified.

Appendix A relaxes Assumption S1; needs and tax liabilities then enter the first-order condition through the marginal utility of consumption. Section 4 restates Assumption S1 in potential-outcomes form (Assumption 2). The first-order condition is

$$z_{it} = n_{it}(g_{it}) (\tau_{it})^\epsilon, \quad \tau_{it} \equiv 1 - T'_t(z_{it}; Q_{it}),$$

where τ_{it} is the marginal net-of-tax rate. Taking logs and first differences yields the estimable equation

$$(2) \quad \Delta \ln z_{it} = \epsilon \Delta \ln \tau_{it} + \Delta \ln n_{it}(g_{it}),$$

where $\Delta \ln n_{it}(g_{it})$ captures changes in potential income, including any direct effect of the event. The parameter ϵ is the estimand: the elasticity of (taxable or work) income with respect to the marginal net-of-tax rate.⁸ Section 3.3 makes precise what it measures for individuals in couples.

3.2 Simulated tax instruments

Since the marginal tax rate is a function of income, $\Delta \ln \tau_{it}$ in equation (2) is endogenous: any shock to income moves the individual along the progressive schedule. The literature addresses this endogeneity with *simulated instruments*: mechanical predictions of the tax change, computed by applying a change in the tax environment to predetermined income (Gruber & Saez, 2002); see Saez et al. (2012) for an overview. Designs differ in what changes the tax environment.

Reform-based instruments. In the standard design, the change in the tax environment is a reform, and the instrument applies the post- and pre-reform schedules to pre-reform income,

$$\Delta \ln \tau_{it}^{\text{reform}} = \ln(1 - T_t^{\text{post}}(z_{i,t-k})) - \ln(1 - T_t^{\text{pre}}(z_{i,t-k})).$$

⁸We suppress an individual subscript on ϵ in this section for readability. Section 4 treats the elasticity as individual-specific (ζ_i) and characterises which weighted averages of the individual parameters our estimands recover.

Validity requires the simulated change to be unrelated to potential-income growth conditional on controls,

$$(3) \quad \text{cov}(\Delta \ln \tau_{it}^{\text{reform}}, \Delta \ln n_{it} \mid z_{i,t-k}) = 0,$$

read as parallel income paths absent the reform. The assumption is demanding because exposure to reforms is usually *between*-income (Jakobsen & Sogaard, 2022): the instrument is a function of lagged income, and potential-income growth is related to the level of income through mean reversion and differential secular income trends, the two leading confounds of this literature.⁹ A further concern, developed in Appendix G, is that reform variation can move different individuals’ realised marginal rates in opposite directions—income effects and extensive-margin responses being the leading channels—in which case the IV estimand no longer aggregates behavioural responses with convex weights.

Event-based instruments. We propose to let the change in the tax environment be a family event that changes the household’s *fiscal configuration*: the set of persons whose incomes are pooled in the tax base, and the family quotient dividing it (Section 2.1).¹⁰ Writing $\mathcal{U}_d$ for the tax unit and Q_d for the quotient before ($d = 0$) and after ($d = 1$) the event—so that d coincides with the event indicator g of Section 3.1—per-part taxable income in year t is

	Event	$d = 0$		$d = 1$	
		$\mathcal{U}_0$	Q_0	$\mathcal{U}_1$	Q_1
$b_{it}(d) = \frac{\sum_{j \in \mathcal{U}_d} z_{jt}}{Q_d},$	Child’s fiscal exit: one dependent child	$\{i, p\}$	2.5	$\{i, p\}$	2
	two dependent children	$\{i, p\}$	3	$\{i, p\}$	2.5
	Marriage (childless couple)	$\{i\}$	1	$\{i, p\}$	2

The household’s pre-event constellation—the number of dependent children, single or joint filing—selects the relevant row and is what Section 4 records as the event type e_i . The

⁹See Jakobsen and Sogaard (2022) for a detailed discussion of equation (3) and of these two channels.

¹⁰Children are part of the parents’ tax unit through age 17 automatically; young adults can remain fiscally attached up to age 21 (25 if in education). Upon detachment, parents may instead deduct support payments (*pension alimentaire*) up to a cap. These choice margins mean the timing of the fiscal exit is not fully mechanical; Appendix B probes manipulation directly with density tests, and pre-trends provide a further check.

instrument is the change in the log net-of-tax rate predicted by the event alone, obtained by applying the pre-event year’s schedule to pre-event incomes under the two configurations:

$$(4) \quad Z_i \equiv \ln(1 - T'_{m_i-1}(b_{i,m_i-1}(1))) - \ln(1 - T'_{m_i-1}(b_{i,m_i-1}(0))),$$

where m_i is the calendar year of the event; for the fiscal exit, m_i is the year in which the first-born turns 21, fixed by fertility two decades prior (Section 3.1). By the common-marginal-rate property of Section 2.1, Z_i is expressed in the units of the household net-of-tax rate τ^H , the first-stage outcome of Section 4. Whether the event is predicted to change a household’s marginal rate, and by how much, depends on where predetermined income sits relative to the thresholds of the post-event schedule—for given own income, on partner income (Figure 1). This is *within*-income variation:¹¹ among individuals with the same own income, the same pre-event bracket, and an event of the same type in the same year, some households are pushed over a bracket threshold and others are not, depending on the partner’s income. At the child’s fiscal exit, thresholds contract and the marginal rate weakly rises at every level of household income, so $Z_i \leq 0$: the event predicts tax increases only. At marriage, the tax base pools and predicted increases and decreases coexist. Table 1 illustrates.

Table 1: Two stylised couples in the year before the first-born turns 21

	Woman A	Woman B
Own income	20,000	20,000
Partner income	35,000	40,000
Bracket at $Q = 2.5$	14%	14%
Predicted bracket at $Q = 2$	14%	30%
Instrument Z_i (equation (4))	0	−0.206

Notes: The table shows two households identical in the woman’s own income and pre-event tax bracket, whose predicted tax treatment upon the first-born’s fiscal exit differs because of partner income. Brackets refer to the 2015 schedule; $-0.206 = \ln(0.70/0.86)$.

The validity requirement for the event-based instrument parallels equation (3): the predicted change Z_i must be unrelated to potential-income growth among comparable households,

$$(5) \quad \text{cov}(Z_i, \Delta \ln n_{it} \mid z_{i,m_i-1}, m_i, e_i) = 0.$$

¹¹Tax variation among taxpayers at the same income level is showcased by H. J. Kleven and Schultz (2014), whose Danish reforms create “variation even for taxpayers at the same income level”; Jakobsen and Søgaaard (2022) develop the associated identifying assumption of constant trend differentials and its graphical validation.

The comparison with the reform design clarifies why the event variation is better placed to meet it. Exposure to reforms is between-income: the instrument varies with the level of lagged own income, precisely the dimension along which mean reversion and secular income trends operate. Exposure to the events is within-income: conditional on own income, the event year, and the event type, Z_i varies only with partner income, so the confounds tied to the level of own income are held fixed by design, and the identifying restriction concerns partner income within narrow bands. Events, unlike reforms, have direct effects on potential income— n_{it} depends on g_{it} —but predicted controls experience the same event, so these direct effects difference out within cells. Section 4 states the within-band restriction as Assumption 1—conditional common trends within cells—and characterises which averages of the individual elasticity parameters the resulting estimands recover.

3.3 A household model

Equation (2) is written for one person. Pairing it with marginal rates produced by a household tax system is standard practice,¹² but under family taxation it leaves the interpretation of ϵ imprecise: partner income enters τ_{it} , and the partner responds to the same rate. We therefore extend the model to a household choosing both spouses’ incomes against a joint budget constraint,

$$\max_{z_{it}, z_{pt}} c_t - \Psi(z_{it}, z_{pt}; n_{it}, n_{pt}) \quad \text{s.t.} \quad c_t \leq z_{it} + z_{pt} - T_t^H(z_{it} + z_{pt}; Q_{it}),$$

where z_{pt} is the partner’s income, $T_t^H(z_{it} + z_{pt}; Q_{it}) = Q_{it} T_t((z_{it} + z_{pt})/Q_{it})$ is the household tax function of equation (1), and Ψ is a convex joint cost of earning income. The joint cost may be additively separable across spouses—the specification underlying most quantitative work on couples taxation (Bick & Fuchs-Schündeln, 2018)—or non-separable, reflecting leisure complementarities, home production, or coordinated retirement (Gelber, 2014; Goux et al., 2014).¹³

¹²Most prominently, H. J. Kleven and Schultz (2014) estimate the one-person model on Danish administrative data in which, despite individual filing, “for a married person, income tax liability depends on both individual incomes and on spousal incomes” (their footnote 6), owing to exemptions transferable across spouses; their tax simulator accounts for this jointness. The couples dimension is incidental to their reform-based variation, whereas it is our identifying variation, which is why we make the household structure explicit.

¹³Quasi-linearity in household consumption carries Assumption S1 to the household: it shuts down income effects, so that neither the partner’s net income nor the change in the household’s tax *liability* affects earnings

Because the household tax base pools incomes, the marginal rate is common to both spouses (Section 2.1), $\partial T_t^H / \partial z_{it} = \partial T_t^H / \partial z_{pt}$, and we write $\tau_{it}^H \equiv 1 - T_t'((z_{it} + z_{pt})/Q_{it})$ for the common net-of-tax rate, an equilibrium object jointly determined with both spouses' choices. The first-order conditions, $\partial \Psi / \partial z_{it} = \partial \Psi / \partial z_{pt} = \tau_{it}^H$, log-linearise to

$$(6) \quad \Delta \ln z_{it} = \epsilon_{ii} \Delta \ln \tau_{it} + \epsilon_{ip} \Delta \ln \tau_{pt} + \Delta \ln n_{it},$$

where τ_{it} and τ_{pt} are the spouse-specific marginal net-of-tax rates and the first subscript of each elasticity indexes whose earnings respond, the second whose rate changes: ϵ_{ii} is the own-rate elasticity of i 's earnings, and ϵ_{ip} is the cross-rate elasticity—the response of i 's earnings to a change in the *partner's* rate, transmitted, under S1, entirely through the partner's behavioural adjustment. The analogous equation for $\Delta \ln z_{pt}$ carries ϵ_{pp} and ϵ_{pi} . A non-zero cross-rate elasticity requires non-separable effort costs, $\partial^2 \Psi / \partial z_{it} \partial z_{pt} \neq 0$.

What matters for interpretation is whether the partner's rate changes along with one's own. Under joint filing the two rates coincide period by period, so any event-induced change is common to both spouses, $\Delta \ln \tau_{it} = \Delta \ln \tau_{pt} = \Delta \ln \tau_{it}^H$, and (6) collapses to

$$\Delta \ln z_{it} = \underbrace{(\epsilon_{ii} + \epsilon_{ip})}_{\equiv \epsilon_i^H} \Delta \ln \tau_{it}^H + \Delta \ln n_{it}.$$

For couples, the coefficient ϵ in equation (2) is therefore to be read as $\epsilon_i^H = \epsilon_{ii} + \epsilon_{ip}$ whenever both spouses' rates move together—as they always do at the child's fiscal exit: the *total* elasticity of own earnings with respect to the household net-of-tax rate, including the component transmitted through the partner's response. At marriage, by contrast, the two spouses' rate *changes* generally differ—the pre-marriage rates were individual—and in some cells the partner's rate does not change at all; we exploit this below. The income-weighted sum of ϵ_i^H across spouses is the household taxable-income elasticity relevant for revenue and Laffer calculations, however the response splits across own- and cross-rate channels. We estimate equation (2) separately by gender, allowing ϵ_i^H to differ across spouses.

choices, and it makes the intra-household allocation of consumption irrelevant for earnings: with transferable utility, any sharing rule supported by lump-sum transfers leaves the first-order conditions unchanged. For the same reason the model's predictions for earnings coincide with those of an efficient collective household à la Chiappori with quasi-linear private utilities—we need not take a stance on the household decision process. See Appendix A for the extension with income effects.

Existing evidence suggests that cross-rate elasticities are small: exploiting the Swedish move towards individual taxation, in which spouses' marginal rates moved independently, Gelber (2014) estimates compensated cross-rate elasticities of approximately 0.05, an order of magnitude below typical own-rate estimates. The separable specification—equivalently, the one-person model applied spouse by spouse—is therefore a good approximation. We note that the distinction is not specific to our design: whenever a reform changes both spouses' marginal rates, reform-based estimates likewise recover combinations of own- and cross-rate responses, a point the literature has left implicit. Moreover, the approximation can be dispensed with: ϵ_{ii} can be isolated (i) from marriage cells in which the partner's rate is unchanged ($\Delta \ln \tau_{pt} = 0$), where the reduced form identifies ϵ_{ii} alone, so that the comparison with common-change cells identifies ϵ_{ip} ; or (ii) by imputing $\hat{\epsilon}_{ip}$ from external estimates.¹⁴

Two extensions of the model, developed in Appendix A, are used below. Allowing for income effects—a concave utility of consumption—adds a virtual-income term to equation (2); absent income effects, compensated and uncompensated elasticities coincide, while in their presence the estimands recover a total response to the event-induced change in the schedule rather than either structural elasticity, and the design's distance-to-threshold variation separately identifies the income effect (Section 4, Assumption 2; Appendix A). And a participation-margin block in the spirit of H. J. Kleven et al. (2009) rationalises the extensive-margin analysis: there, the relevant price is the participation tax wedge, whose first-euro marginal rate is exactly what partner income and the fiscal configuration determine under the quotient system.

¹⁴In more detail, there are two routes from the identified total to the structural own-rate elasticity $\epsilon_{ii} = \epsilon_i^H - \epsilon_{ip}$. **Route 1** (*spousal separability, S2*): assume $\partial^2 \Psi / \partial z_{it} \partial z_{pt} = 0$, so that $\Psi = \psi(z_{it}; n_{it}) + \psi(z_{pt}; n_{pt})$. Then $\epsilon_{ip} = 0$ and $\epsilon_i^H = \epsilon_{ii}$; specialising ψ to the iso-elastic form, the household problem separates into two copies of the one-person model, each facing the household rate—the formal justification for the standard practice of pairing the one-person model with household-rate variation. **Route 2** (*external adjustment*): keep $\epsilon_{ip} \neq 0$ and impute it from Gelber (2014); the structural elasticity is recovered as $\epsilon_{ii} = \epsilon_i^H - \hat{\epsilon}_{ip}$, reported as an adjustment or as bounds. Three remarks discipline the latter exercise. The sign: leisure complementarity implies $\epsilon_{ip} > 0$, so the unadjusted ϵ_i^H *overstates* the structural elasticity. The magnitude: plausible values of ϵ_{ip} are an order of magnitude below our estimates, so the adjustment is second-order for our conclusions. The scope: external estimates are compensated, gender-specific, and from a different population and margin, so we treat them as orders of magnitude rather than point corrections, and apply the adjustment to the intensive margin only—for participation, external cross-elasticity estimates are scarce and the added-worker channel can reverse the sign. The distinction matters most for the heterogeneity results: gender differences in ϵ_i^H are preference parameters only under Route 1, or after the Route 2 adjustment; without either, a larger measured response of one spouse may partly reflect the other spouse's adjustment transmitted within the household. Because we estimate both spouses' responses to the same household shock within the same cells, the pair of estimates is itself informative about such reallocation, and we return to this when interpreting the heterogeneity results.

4 Empirical strategy

For each individual i experiencing the event in calendar year m_i , we construct an event window around m_i , with event time $r \equiv t - m_i$, a pre-period ($r = -3, -2, -1$) and a post-period ($r = 0, \dots, R$); Figure 3 summarises the design (the sample is described below).

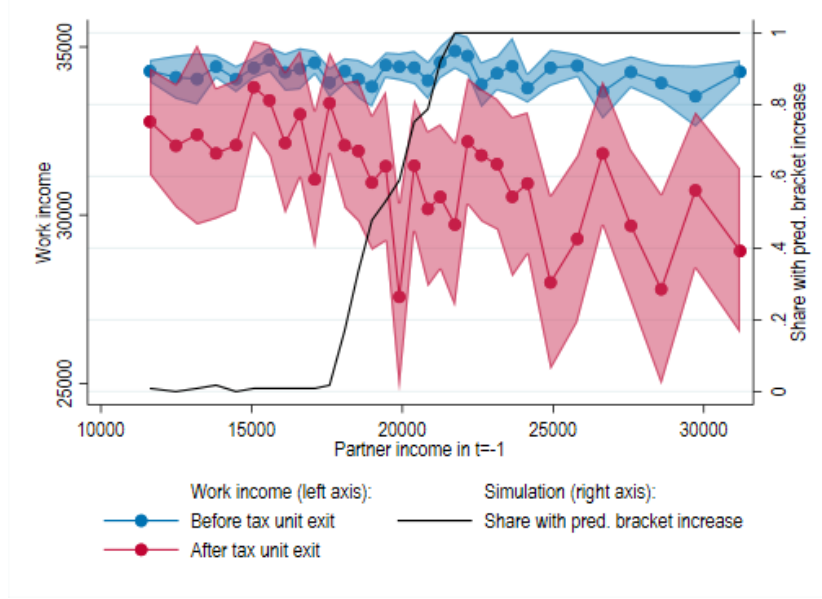


Figure 3: Identifying variation at the first-born's fiscal exit

Notes: Sample: women in married or civil-union tax units whose first-born exits the tax unit between 2003 and 2023. The probability of a predicted tax increase rises sharply with partner income, dividing the sample into a validation region, in which the tax unit experiences the exit but no change in tax bracket, and a treatment region, in which the exit is predicted to raise the marginal tax rate. Identification comes from the difference between the two regions; the validation region provides a direct check that partner income does not by itself predict earnings growth. The figure follows the graphical validation approach of Jakobsen and Sogaard (2022).

Outcomes are measured relative to the pre-event year: earnings growth $\Delta y_{ir} \equiv \ln z_{ir} - \ln z_{i,-1}$ and net-of-tax rate growth $\Delta \text{NTR}_{ir} \equiv \ln \tau_{ir}^H - \ln \tau_{i,-1}^H$, with τ^H the household net-of-tax rate of Section 3.3. A possible estimand for the elasticity of earnings is the difference-in-differences estimand that compares the earnings and net-of-tax rate growth of treated and control households:

$$\beta^{DD} \equiv \frac{\overbrace{\mathbb{E}[\Delta y \mid Z \neq 0] - \mathbb{E}[\Delta y \mid Z = 0]}^{\equiv RF \text{ (Earnings DiD)}}}{\underbrace{\mathbb{E}[\Delta \text{NTR} \mid Z \neq 0] - \mathbb{E}[\Delta \text{NTR} \mid Z = 0]}_{\equiv FS \text{ (Net-of-tax rate DiD)}}}.$$

Two concerns arise with this estimand. First, Z takes many values: pooling compares households experiencing tax changes of different sizes, which can place negative weights on individual responses when elasticities are heterogeneous (Callaway et al., 2024; Graber et al., 2026). Second, treated and control households differ systematically: for given own income, it is higher partner income that pushes household income across a threshold of the post-event schedule, so the pooled comparison mixes the tax response with any differential earnings growth associated with partner income. We address both concerns by conducting the analysis at the cell–dose level. Let $X_i \equiv (z_{i,-1}, z_{p,-1}, m_i, e_i)$ collect own income, partner income, the event calendar year, and the event type, all predetermined. Cells $\mathcal{I}_c$ partition the support of X into a narrow own-income bin, a partner-income interval, an event year, and an event type; we write $c(X_i)$ for the cell of household i . The pre-event marginal tax bracket is a deterministic function of X_i , so a common baseline bracket is a property of the partition rather than a separate variable.¹⁵ Throughout, expectations subscripted by c , $\mathbb{E}_c[\cdot]$, are taken over households in cell $\mathcal{I}_c$. Within a cell, for each dose δ in the within-cell support of Z , we compare households with $Z_i = \delta$ to households with $Z_i = 0$: holding own income fixed removes the dimension along which incomes trend differentially, and comparing one dose at a time against the zero-dose group avoids comparisons across treatment intensities. Summary parameters aggregate the cell–dose estimates with explicit non-negative frequency weights (Section 4.3); estimation and inference, including a matching implementation, are described in Section 4.4.

To fix ideas, consider one such cell: tax units with pre-event own income between €34,000 and €36,000, a pre-event marginal rate of 14%, and 2.5 fiscal parts—a married or civil-union couple whose one dependent child exits the tax unit at $r = 0$. Within this cell, households whose partner earns little enough that per-part income stays below the threshold once the quotient falls to 2 have $Z_i = 0$; households with higher-earning partners are pushed into the 30% bracket and have $Z_i = \delta < 0$. Table 2 reports descriptive statistics for the two groups and Figure 4 plots their earnings paths around the event.

¹⁵The instrument is likewise a deterministic function of the predetermined covariates: given own income, partner income, the event year, and the event type, equation (4) pins down Z_i . Within a cell, variation in Z_i therefore reflects only variation in partner income over the cell’s interval.

Table 2: Predicted-treated and predicted-control households in one cell

	Women		Men	
	Pred. control (N=2,502)	Pred. treated (N=2,008)	Pred. control (N=5,405)	Pred. treated (N=2,809)
Work income	34,917	34,920	34,951	35,024
Partner's work income	23,672	30,436	21,220	28,944
Age	52.0	53.0	53.1	54.1
Partner's age	53.3	54.3	51.4	52.7
Predicted tax change	0.00	0.16	0.00	0.16
Actual tax change	0.01	0.10	0.01	0.10

Notes: The cell comprises tax units with pre-event own income in $[\text{€}34,000, \text{€}36,000]$, a pre-event marginal tax rate of 14%, and 2.5 fiscal parts, whose first-born exits the tax unit at $r = 0$. Predicted controls are households for which the event is not predicted to change the marginal tax rate ($Z_i = 0$); predicted treated are households pushed into the 30% bracket. All variables are measured in the pre-event year $r = -1$, except the predicted and actual tax changes, which are measured between $r = -1$ and $r = 1$. The two groups are balanced on own income and close in age, and differ in partner income, which is what determines whether the fall in the family quotient pushes per-part income across the threshold of the post-event schedule. The predicted change is the instrument of equation (4); the actual change is the realised change in the household net-of-tax rate.

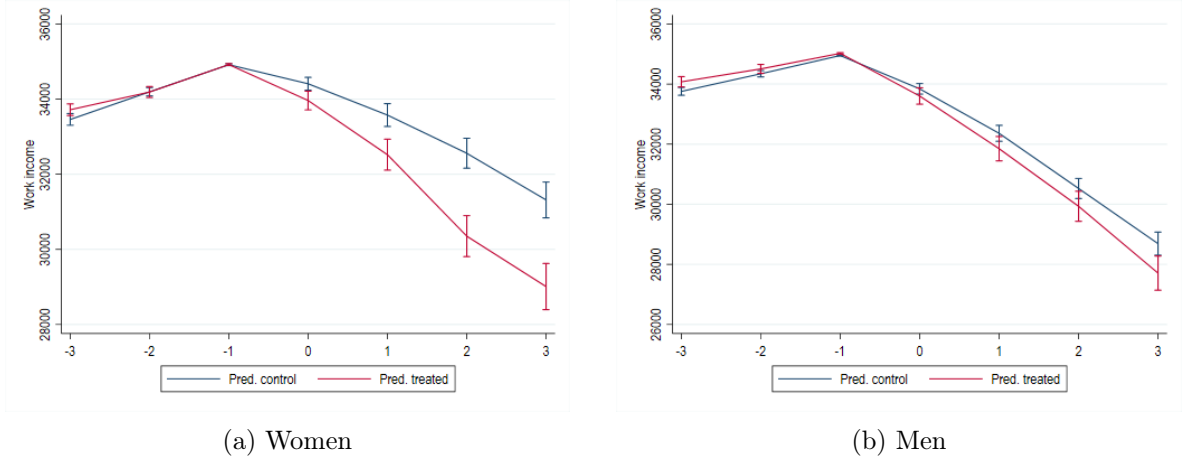


Figure 4: Earnings around the first-born's fiscal exit, by predicted treatment status

Notes: Raw means, without matching or covariate adjustment. Labour income is measured relative to the pre-event year $r = -1$; the vertical line marks the event. The series distinguish households predicted to change tax bracket at the event ($Z_i = \delta < 0$) from those predicted not to ($Z_i = 0$).

4.1 Potential earnings model

To consider the causal interpretation of these comparisons, we introduce a potential earnings model that links them to individual elasticity parameters.

Let $D_{ir} \in \{0, 1\}$ denote the fiscal configuration the household is taxed under at event time r ; mechanically, $D_{ir} = \mathbb{1}[r \geq 0]$, and in the notation of Section 3, $D_{ir} = g_{i, m_i + r}$. Given the two configurations, we let $Y_{ir}(d)$ denote potential log earnings at event time r under configuration d , and $\text{NTR}_{ir}(d)$ the corresponding potential log net-of-tax rate.¹⁶ The potential outcomes map to observed outcomes through

$$(7) \quad Y_{ir} = Y_{ir}(0) + D_{ir} \mathbb{1}[Z_i \neq 0] [Y_{ir}(1) - Y_{ir}(0)], \quad \text{NTR}_{ir} = \text{NTR}_{ir}(0) + D_{ir} \mathbb{1}[Z_i \neq 0] [\text{NTR}_{ir}(1) - \text{NTR}_{ir}(0)].$$

For the predicted-control group, observed outcomes are thus untreated potential outcomes at all horizons: the mapping embeds the restriction that any tax change they experience is not event-led. We gauge this restriction directly with the control group's realised net-of-tax rate changes (Appendix B).

Following Saez et al. (2012) and Graber et al. (2026), we assume the potential earnings function is

$$(8) \quad Y_{ir}(\text{NTR}, d) = \zeta_i \times \text{NTR} + \nu_{ir}(d),$$

where ζ_i is the individual's elasticity parameter—the total household-rate elasticity ϵ_i^H of Section 3.3—and the $\nu_{ir}(d)$ can vary freely across individuals and event times. The term $\nu_{ir}(d)$ collects all determinants of log earnings other than the marginal net-of-tax rate, and its two indices play distinct roles. The event-time index r carries everything that evolves over the event window irrespective of the tax formula: life-cycle and calendar-time earnings growth and, in particular, the direct, non-tax consequences of the event itself—changed time use, home production, transfers. Because both potential outcomes are defined with the event having occurred, these direct effects are common to the two configurations by construction; writing

$$(9) \quad \nu_{ir}(d) = \alpha_{ir} + \eta_{ir}(d), \quad \eta_{ir}(0) \equiv 0,$$

¹⁶ $Y_{ir}(0)$ is not a no-event counterfactual: the event occurs under both configurations, and only the fiscal configuration is varied.

they enter through α_{ir} . The configuration argument d carries any non-rate channel through which the fiscal configuration itself could affect earnings—above all, the change in tax liability. Assumption 2 below shuts down this channel.

We impose the following assumptions.

Assumption 1 (Conditional common trends). *For every cell, horizon r , and dose δ in the within-cell support of Z ,*

$$\mathbb{E}_c[\Delta_r Y(0) \mid Z_i = \delta] = \mathbb{E}_c[\Delta_r Y(0) \mid Z_i = 0], \quad \mathbb{E}_c[\Delta_r \text{NTR}(0) \mid Z_i = \delta] = \mathbb{E}_c[\Delta_r \text{NTR}(0) \mid Z_i = 0].$$

Assumption 1 conditions on the cell and contrasts the dose, in the form familiar from conditional difference-in-differences (Abadie, 2005; Callaway & Sant’Anna, 2021; Heckman et al., 1997). It leaves untreated growth unrestricted *across* cells: it may vary freely with own income, the partner-income interval, the event year, the event type, and the baseline bracket. Assumption 1 holds, in particular, whenever untreated growth does not vary within cells at all, $\mathbb{E}[\Delta_r Y(0) \mid X_i = x] = f_y^r(c(x))$ for all x , and likewise for $\Delta_r \text{NTR}(0)$: since own income, the event year, and the event type are fixed by the cell, this says that the exact position of partner income within the cell’s partner-income band does not predict untreated growth, and it implies Assumption 1 because the dose groups within a cell differ only in where partner income falls within the band. By the potential earnings function and (9), $\Delta_r Y(0) = \zeta_i \Delta_r \text{NTR}(0) + \Delta_r \alpha_{ir}$, so the restriction concerns untreated rate drift and α -growth—the latter including the direct effects of the event.¹⁷

Assumption 2 (Exclusion restriction). $\nu_{ir}(0) = \nu_{ir}(1)$ with probability one; equivalently, $\eta_{ir}(1) = 0$ in (9).

The configuration change affects earnings only through the marginal net-of-tax rate. It is Assumption S1 of Section 3.1 in potential-outcomes form.¹⁸ Following Graber et al. (2026), Assumption 2 holds if and only if there are no income effects, in which case ζ_i is the compensated elasticity, which coincides with the uncompensated one (their Proposition 3). If

¹⁷Narrower partner-income bands weaken the requirement, and the nearest-neighbour matching implementation weakens it further by comparing households just either side of the predicted threshold; pre-event trends provide a direct test in every cell.

¹⁸In terms of (9), Assumption 2 restricts only the configuration-specific component, $\eta_{ir}(1) = 0$; the direct effects of the event enter through α_{ir} , which carries no configuration argument and remains unrestricted.

income effects are present, the estimands recover neither the compensated nor the uncompensated elasticity but a total response to the event-induced change in the tax schedule (their Proposition 4). The income effect, however, is proportional to the amount of income that the configuration change shifts across a bracket threshold—a quantity that can be instrumented in the same way as the marginal rate, by applying the pre-event year’s schedule to pre-event incomes under the two configurations as in equation (4). Appendix A uses this second, event-based instrument to separately identify income effects and illustrates the argument graphically.

Assumption 3 (Monotonicity). $Z_i [\text{NTR}_{ir}(1) - \text{NTR}_{ir}(0)] \geq 0$ with probability one, for all $r \geq 0$.

The condition is stated on the event-induced component of rates, $\Phi_{ir} \equiv \text{NTR}_{ir}(1) - \text{NTR}_{ir}(0)$: among households predicted a rate increase ($Z_i < 0$), the configuration change does not raise any household’s net-of-tax rate, and conversely for $Z_i > 0$. It restricts only variation caused by the event; realised rates may move in either direction through untreated drift, $\Delta_r \text{NTR}(0)$, without violating it. Dose by dose, this is the standard no-defiers condition ensuring that Wald estimands reflect positively weighted averages of individual parameters (Angrist & Imbens, 1995; Imbens & Angrist, 1994). Appendix G shows that the analogous condition is easily violated in reform settings, where income effects and extensive-margin responses can move earnings across brackets against the reform’s prediction. A configuration change shifts the thresholds of the schedule rather than its gradient: a violation requires the household’s joint income to respond so strongly that per-part income re-crosses the shifted threshold, and partner income places a floor under this adjustment. The appendix formalises the argument with a simulation.

4.2 The causal interpretation of the estimands

For each cell and dose δ , the reduced-form and first-stage contrasts are

$$RF_c^r(\delta) \equiv \mathbb{E}_c[\Delta y_{ir} \mid Z_i = \delta] - \mathbb{E}_c[\Delta y_{ir} \mid Z_i = 0], \quad FS_c^r(\delta) \equiv \mathbb{E}_c[\Delta \text{NTR}_{ir} \mid Z_i = \delta] - \mathbb{E}_c[\Delta \text{NTR}_{ir} \mid Z_i = 0],$$

where $\mathbb{E}_c[\cdot \mid Z_i = \delta] \equiv \mathbb{E}[\cdot \mid Z_i = \delta, i \in \mathcal{I}_c]$, and our headline estimand is the cell-dose elasticity

$$\beta_c^r(\delta) \equiv \frac{RF_c^r(\delta)}{FS_c^r(\delta)},$$

the disaggregated counterpart of β^{DD} ; dividing the reduced form by the predicted change instead, $RF_c^r(\delta)/\delta$, yields the policy-elasticity variant.

Proposition 1 (Causal interpretation). *Let $\Phi_{ir} \equiv \text{NTR}_{ir}(1) - \text{NTR}_{ir}(0)$ denote the individual first-stage response. Under the mapping (7), the potential earnings function (8), and Assumptions 1–3, for every cell, dose $\delta \neq 0$, and horizon $r \geq 0$ with $FS_c^r(\delta) \neq 0$:*

$$\beta_c^r(\delta) = \mathbb{E}_c[\omega_{ir} \zeta_i \mid Z_i = \delta], \quad \omega_{ir} \equiv \frac{\Phi_{ir}}{\mathbb{E}_c[\Phi_{ir} \mid Z_i = \delta]},$$

with weights $\omega_{ir} \geq 0$ averaging to one: the cell-dose elasticity is a positively weighted average of individual elasticity parameters among the predicted treated.

Proof. By (7) and (8), $\Delta y_{ir} = \Delta_r Y_i(0) + D_{ir} \mathbb{1}[Z_i \neq 0] \zeta_i \Phi_{ir}$ and $\Delta \text{NTR}_{ir} = \Delta_r \text{NTR}_i(0) + D_{ir} \mathbb{1}[Z_i \neq 0] \Phi_{ir}$. By Assumption 1, the conditional means of the untreated components are equal for the $Z_i = \delta$ and $Z_i = 0$ groups of the same cell, so they cancel in the contrasts: $RF_c^r(\delta) = \mathbb{E}_c[\zeta_i \Phi_{ir} \mid Z_i = \delta]$ and $FS_c^r(\delta) = \mathbb{E}_c[\Phi_{ir} \mid Z_i = \delta]$. Taking the ratio gives the weighted-average representation; by Assumption 3, all Φ_{ir} in the $Z_i = \delta$ group share the sign of δ , so the weights are non-negative. $\square$

Proposition 1 is the analogue, for event-based instruments, of the characterisation in Graber et al. (2026) of when reform-based ETI estimands are causal. Their two conditions—that the instrument enter as a binary contrast, and that income controls be rich enough to reproduce the conditional mean of the instrument—appear here as, respectively, a feature of the design (each estimand uses the binary contrast $\mathbb{1}\{Z_i = \delta\}$ against the zero-dose group) and the identifying assumption itself (Assumption 1: within cells, the instrument is unrelated to untreated growth).

4.3 Aggregation

Because the aggregation across doses and cells uses explicit non-negative weights, the summary parameters inherit the causal interpretation of Proposition 1. We aggregate in two steps. Within a cell, the dose-specific elasticities are combined with frequency weights over the predicted treated,

$$\beta_c^r \equiv \sum_{\delta \in \mathcal{D}_c} w_c(\delta) \beta_c^r(\delta), \quad w_c(\delta) \equiv \Pr_c(Z_i = \delta \mid Z_i \neq 0),$$

where $\mathcal{D}_c$ denotes the within-cell support of non-zero doses. Across cells, we form group parameters for the paper’s headline dimensions—the own-income group p and the within-couple split s —as

$$\epsilon_{p,s}^r \equiv \sum_{c \in \mathcal{G}_{p,s}} v_{p,s}(c) \beta_c^r,$$

where $\mathcal{G}_{p,s}$ is the set of cells in group (p, s) and $v_{p,s}(c)$ is cell c ’s share of the group’s predicted-treated households. We report $\epsilon_{p,s}^r$ as an event-time profile; the headline estimates of Section 5 fix a reference horizon r^* . Substituting Proposition 1, $\epsilon_{p,s}^r$ is a weighted average of the individual parameters ζ_i among the predicted treated of the group, with weights proportional to $v_{p,s}(c) w_c(\delta) \omega_{ir}$ and hence non-negative. The weights are known sample frequencies rather than estimator-implied: in contrast to pooled regressions across treatment intensities, whose implicit weights can turn negative under heterogeneity (Callaway et al., 2024; Graber et al., 2026), the weighting here is explicit, and alternative policy-relevant weightings can be substituted without affecting the identification argument.

4.4 Estimation and inference

The contrasts of Section 4.2 are estimated by their sample analogues. Within each cell, all doses are estimated jointly against the common zero-dose group by least squares,

$$(10) \quad \Delta y_{ir} = a_c^r + \sum_{\delta \in \mathcal{D}_c} RF_c^r(\delta) \mathbf{1}\{Z_i = \delta\} + u_{ir}, \quad i \in \mathcal{I}_c,$$

and identically for ΔNTR_{ir} , which yields the $FS_c^r(\delta)$; the cell–dose elasticity is the ratio of the two coefficients, numerically identical to two-stage least squares on the cell–dose sample

with instrument $\mathbb{1}\{Z_i = \delta\}$. Because every regressor is a binary contrast against the omitted zero-dose group, the coefficients are exactly the group-mean differences of Section 4.2: no implicit weighting across doses arises, so estimation preserves the binary-contrast condition of Graber et al. (2026). Estimating all doses of a cell in one regression also delivers their joint covariance, which inference must track because the zero-dose households serve as the common comparison group for every dose. Estimating (10) at pre-event horizons $r < 0$ produces the placebo contrasts implied by Assumption 1, which we report alongside the post-event estimates.

The matching implementation tightens the same comparisons: for each treated household, the zero-dose comparison is restricted to nearest neighbours in pre-event partner income within the cell, so that treated and comparison households sit just either side of the predicted threshold (footnote 17); RF , FS and β are then matched differences in means and their ratio.

Standard errors follow from the delta method applied to the jointly estimated (RF, FS) vector of each cell, with variances clustered at the household level across stacked horizons and cells treated as independent; the same delta-method step propagates to the aggregates β_c^r and $\epsilon_{p,s}^r$, whose weights are sample frequencies with negligible estimation error. Cell-dose comparisons with fewer than $\underline{N}$ treated or $\underline{N}_0$ zero-dose households are excluded from estimation and aggregation.

5 Results

This section applies the empirical strategy of Section 4 to the universe of first-born fiscal exits in France between 2003 and 2023.¹⁹ The results are preliminary: the estimates and figures reported below are being re-exported, and both magnitudes and figure formatting are subject to revision.

Estimates are obtained from the cell-level regression of equation (10): within each cell, all doses are estimated jointly against the common zero-dose group, and the reduced form is divided by the corresponding first stage to give the cell-dose elasticity $\beta_c^r(\delta)$. Cell-dose estimates are

¹⁹Marriage, which generates predicted tax changes of both signs, serves as a robustness design; Appendix D reports those estimates. At the fiscal exit the family quotient falls, so the marginal rate weakly rises at every level of household income and $Z_i \leq 0$ throughout; the heterogeneity by direction of the predicted tax change that the marriage design permits is therefore reported there.

aggregated with the explicit non-negative frequency weights of Section 4.3. Throughout, we report estimates at the reference horizon $r = 1$, so that the outcome is earnings growth between the pre-event year and the year following the event, and all analyses are conducted separately by gender. The main text reports intensive-margin responses; extensive-margin estimates, defined on an indicator for positive labour income at $r = 1$, are reported in Appendix E. Event-time profiles, which trace $\beta_c^r(\delta)$ over the window and provide a direct check on the pre-event years, are reported in Appendix F.

Pooling across doses and cells, the intensive-margin elasticity is [X] for women and [X] for men. This aggregate masks substantial heterogeneity, which we document along two dimensions: own income (Section 5.1) and the composition of household income (Section 5.2).

5.1 Heterogeneity by own income

Figure 5 reports elasticities by own pre-event income, aggregating the cell-dose estimates within own-income groups. For women, the estimates decline with income: responses are largest at the bottom of the distribution and attenuate as own income rises.

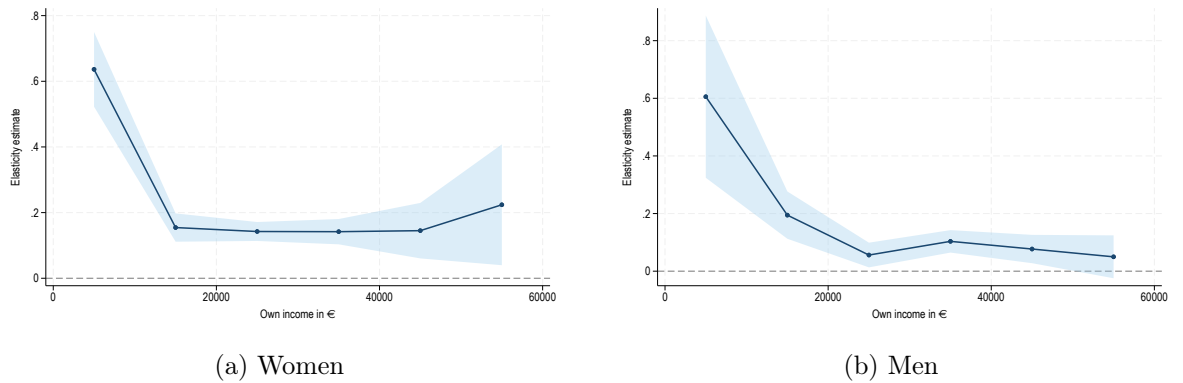


Figure 5: Intensive-margin elasticities along the own-income distribution

Notes: Sample: individuals in married or civil-union tax units whose first-born exits the tax unit between 2003 and 2023. The outcome is earnings growth between $r = -1$ and $r = 1$. Estimates are obtained as described in Section 4: cell-dose elasticities from the regression of equation (10) are aggregated within own-income groups using the frequency weights of Section 4.3. The horizontal axis is own pre-event income, measured at $r = -1$. Shaded areas / bars are [TODO: 95% confidence intervals].

5.2 Heterogeneity by household income composition

The own-income profile aggregates over households that differ in how income is divided within the couple. Figure 6 disaggregates it, reporting elasticities over own *and* partner pre-event income. The horizontal axis is own income, the vertical axis partner income; the diagonal separates households in which the individual is the primary earner from those in which the partner is.

The map shows that the negative own-income gradient of Figure 5 is not driven by the households with the lowest household income. Responses are largest among individuals with low own income *and* a substantially higher-earning partner—secondary earners in couples where the partner carries most of the household’s income—rather than among those whose partner also earns little.

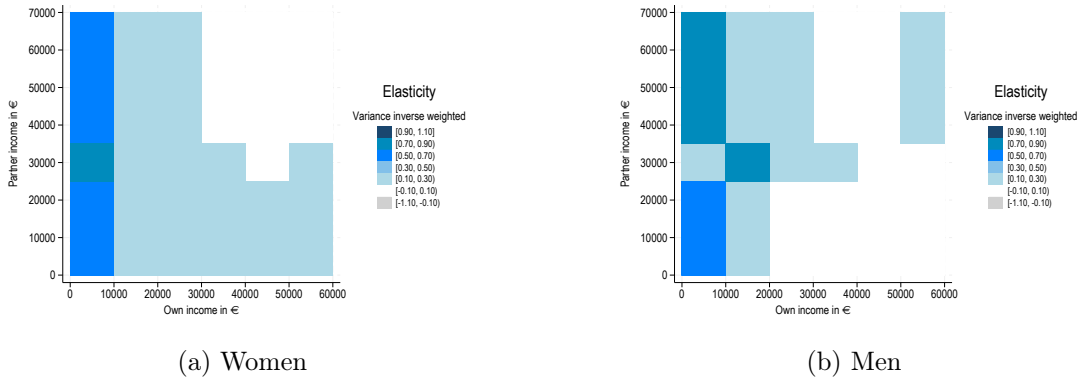


Figure 6: Intensive-margin elasticities by own and partner income

Notes: Sample: individuals in married or civil-union tax units whose first-born exits the tax unit between 2003 and 2023. The outcome is earnings growth between $r = -1$ and $r = 1$. Estimates are obtained as described in Section 4: cell-dose elasticities from the regression of equation (10) are aggregated within groups defined by own and partner income, using the frequency weights of Section 4.3. Both incomes are measured at $r = -1$. Cells with fewer than $\lfloor N \rfloor$ observations are left blank.

6 On the relevance of accounting for heterogeneity

Section 5 documented substantial heterogeneity in estimated elasticities, both across the income distribution and by the composition of income within the household. In this section, we examine whether this heterogeneity matters for tax design.

We assess this question against two methodologically distinct benchmarks. The first is the normative optimum: the Mirrlees–Saez tradition solves for the tax schedule that maximises a social welfare function subject to resource and incentive constraints (Mirrlees, 1971; Saez, 2001), and its counterpart for couples solves the corresponding multidimensional screening problem (Golosov & Krasikov, 2025; H. J. Kleven et al., 2009). The second is the evaluation of reforms: the perturbation approach of Bierbrauer et al. (2024) takes the observed tax system as given and derives sufficient statistics under which a move towards more individualised taxation of couples is Pareto- or welfare-improving. Section 6.1 conducts the first exercise using our estimates of tax elasticities by own income; Section 6.2 conducts the second using our results on heterogeneity in tax responsiveness by the division of income within couples.

6.1 Optimal tax accounting for heterogeneity across income

The modern nonlinear-income-tax framework originates with Mirrlees (1971). A government chooses a tax schedule to maximise a social welfare function subject to its resource constraint and to incentive compatibility; because ability is unobserved, the schedule can condition on earnings but not directly on ability. Saez (2001) expresses the resulting optimality conditions in terms of empirically meaningful sufficient statistics: behavioural elasticities, the earnings distribution, and social marginal welfare weights. Importantly for our purposes, these conditions are local. The optimal marginal rate at earnings level z balances the mechanical revenue and welfare effects of a small rate change against the behavioural response of taxpayers at z , which is governed by the elasticity *at* z . Nothing in the formula requires this elasticity to be common across the distribution; the constant-elasticity calibrations used in practice are a convenience rather than a feature of the theory.²⁰

We adapt the sufficient-statistics implementation of Brewer et al. (2010) and replace its common elasticity with a profile $e(z)$. Figure 7 provides an illustrative comparison. The dashed line assumes a constant elasticity of 0.25; the solid line instead uses a smooth elasticity profile that declines from 0.4 at low earnings to 0.1 at high earnings. Under both specifications the schedule is U-shaped over the displayed range. The comparison isolates the pointwise inverse-

²⁰The sufficient-statistics formula of Saez (2001) already allows the elasticity to vary with earnings. We evaluate $e(z)$ pointwise while holding the earnings distribution, the social marginal welfare weights, and the revenue requirement fixed. The current figure uses an illustrative smooth profile declining from 0.4 to 0.1; in the final application, this profile will be replaced by an interpolation of our estimated income-group elasticities.

elasticity logic of the formula: the higher elasticity at the bottom lowers optimal marginal rates there, while the lower elasticity higher in the distribution raises them, so the heterogeneous-elasticity schedule is substantially steeper than the constant-elasticity benchmark. A single elasticity calibrated to an economy-wide average accordingly misstates optimal rates at both ends of the distribution.

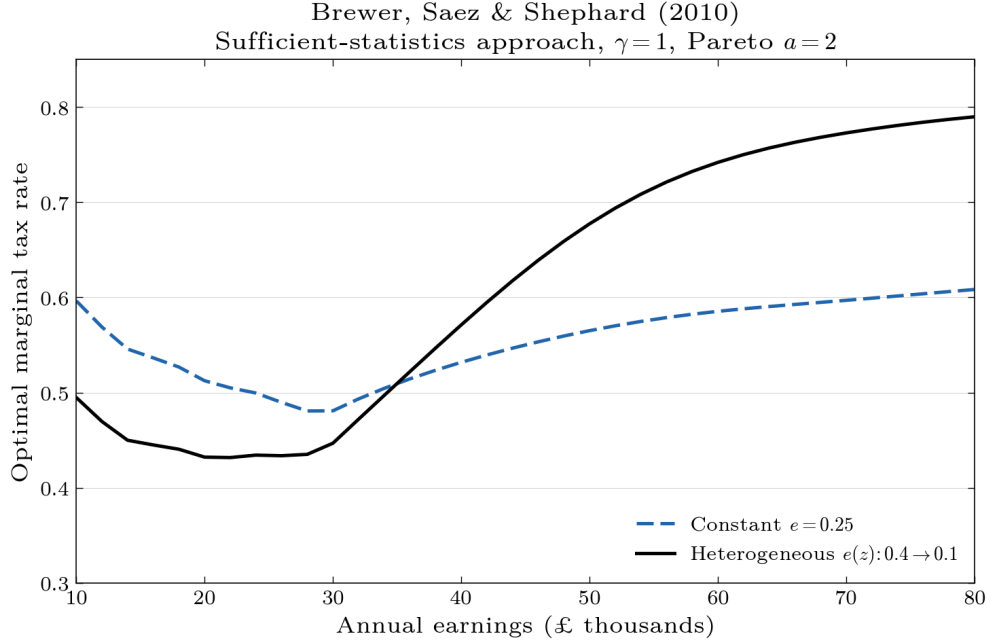


Figure 7: Optimal marginal tax rates under constant and heterogeneous elasticities

Notes: The figure plots optimal marginal tax rates from the sufficient-statistics implementation of Brewer et al. (2010). The dashed line assumes a constant elasticity of 0.25, while the solid line uses an illustrative smooth profile declining from 0.4 to 0.1. The calibration uses the UK earnings distribution, logarithmic social welfare ($\gamma = 1$), a Pareto-tail parameter of $a = 2$, and a revenue requirement of £4,000 per capita.

This exercise exploits only the variation of responses with own earnings. By construction, a one-dimensional formula has no notion of a partner and therefore cannot absorb the variation of elasticities with the composition of household income. Using that dimension requires a framework for couples, to which we now turn.

6.2 Assessment of jointness: accounting for heterogeneity by within-couple split

For couples, solving for the optimal two-dimensional tax schedule is a multidimensional screening problem, which is analytically demanding even under strong simplifying assumptions (Goloso & Krasikov, 2025; H. J. Kleven et al., 2009). In particular, the analytical characterisation in Goloso and Krasikov (2025) relies on a constant elasticity that is common to all individuals and therefore symmetric across spouses; its results would not survive elasticities that vary with the division of earnings within the couple, an extension that is, to our knowledge, work in progress. We therefore follow Bierbrauer et al. (2024), who evaluate reforms of the *existing* system rather than solving for the optimum. Starting from status-quo couple taxation, they derive sufficient statistics that determine whether reforms towards more individualised taxation—reforms that allow the marginal rates of the two spouses to differ, lowering them on secondary relative to primary earnings in a revenue-neutral way—are Pareto-improving and, where they are not, whether they raise welfare under given welfare weights.

Their baseline calibration sets the elasticity of primary earnings to 0.25 and that of secondary earnings to 0.75. Our cell-specific estimates $\epsilon_{p,s}$ replace these two constants with an estimated surface: responses vary with the couple’s position in the joint distribution and with the division of earnings within it.

Figure 8 displays the welfare gains from moving towards a more individualised tax system under the constant-elasticity calibration (left column) and under our cell-specific estimates (right column). The first row assigns equal welfare weights to all households; the second uses near-Rawlsian weights that place greater weight on low-income households.

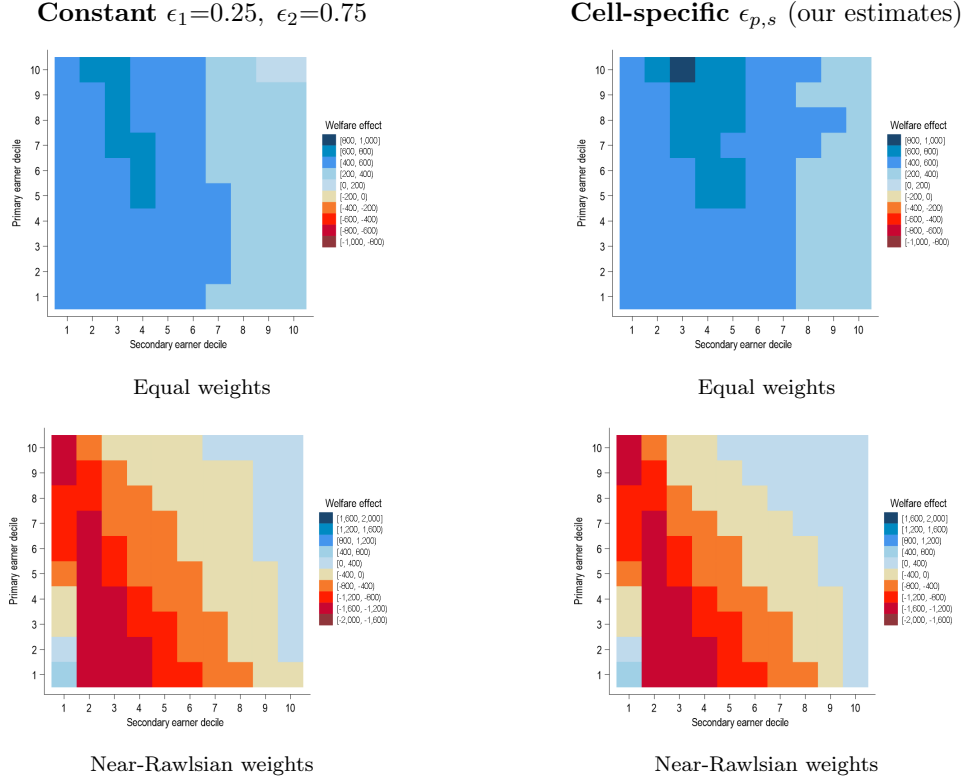


Figure 8: Welfare gains from a more individualised tax system

Notes: The left column uses constant elasticities of $\epsilon_1 = 0.25$ for primary earners and $\epsilon_2 = 0.75$ for secondary earners. The right column uses cell-specific elasticity estimates $\epsilon_{p,s}$. The top row assumes equal welfare weights, while the bottom row assumes near-Rawlsian welfare weights that place greater weight on low-income households. Comparing columns within each row shows that the assessment is sensitive to the interaction between the location of large elasticities in the joint distribution and the welfare weights: the cell-specific estimates raise the assessed gains from individualisation, most markedly under near-Rawlsian weights.

The two exercises show that accounting for heterogeneity changes the shape of the optimal tax schedule and the assessment of reforms towards more individual taxation of couples.²¹²²²³

²¹The direction of the optimal primary–secondary differential is itself unsettled. With a common elasticity for all individuals, optimal marginal rates on secondary earners are on average *higher* than on primary earners, because low secondary earnings identify couples to whom a redistributive planner wants to target transfers (Goloso & Krasikov, 2025); the classic argument of Boskin and Sheshinski (1983) instead prescribes *lower* rates on the more responsive spouse. Which force dominates depends on the size and location of the elasticity gap between spouses—the object we estimate.

²²Constant-elasticity calibrations are not an innocuous simplification in one dimension either: the welfare assessment of the post-war US tax cuts reverses sign as the assumed constant elasticity of taxable income is varied over plausible values (Bierbrauer et al., 2021); estimated elasticity profiles remove this ambiguity.

²³A further margin, which we do not pursue, is political feasibility: with heterogeneous elasticities, assessments of the political feasibility of moves towards individualisation may change (Bierbrauer et al., 2025).

7 Conclusion

This paper proposes a *within*-income source of identifying variation for estimating tax elasticities, suited to mapping heterogeneity across the income distribution and within couples in a way reform- and bunching-based designs are not. Exploiting the institutional features of French family taxation, we use the fiscal exit of a couple’s first-born child—and, as a robustness design, marriage—as an event that changes the mapping from incomes to marginal tax rates, generating variation in marginal incentives through not-yet-tax-liable partner income under transparently testable assumptions.

Using the universe of French tax declarations from 2003 to 2023 and a panel structure that follows individuals across household-unit changes, we document substantial heterogeneity in tax responsiveness by own income, by gender, and by the division of income within couples. Elasticities decline with income and are shifted upward for women; the gradient is driven not by the households with the lowest joint income, but by individuals with low own income and a substantially higher-earning partner.

These estimates are not a refinement of a known number: feeding them into standard frameworks for tax design changes the conclusions those frameworks reach. The optimal tax schedule implied by the sufficient-statistics formula is substantially steeper than under a constant-elasticity benchmark, and the assessed welfare gains from reforms towards more individual taxation of couples are larger once the constant elasticities such evaluations conventionally assume are replaced with our estimates (Section 6).

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Appendix A Model extensions

Income effects

The quasi-linearity of household consumption (Assumption S1) shuts down income effects. Replacing linear consumption with a concave utility $v(c_t)$, the household problem becomes

$$\max_{z_{it}, z_{pt}} v(c_t) - \Psi(z_{it}, z_{pt}; n_{it}, n_{pt}) \quad \text{s.t.} \quad c_t \leq z_{it} + z_{pt} - T_t^H(z_{it} + z_{pt}; Q_{it}) - \kappa_{it}(g_{it}),$$

where $\kappa_{it}(g_{it})$ collects lump-sum household expenses, which may vary with the event (expenses for the child). The first-order conditions become

$$\frac{\partial \Psi}{\partial z_{jt}} = v'(c_t) \tau_{jt}^H, \quad j \in \{i, p\} :$$

earnings choices now depend on the marginal utility of household consumption, and therefore on partner income, on the level of the tax *liability*, and on needs κ_{it} . Log-linearising adds a virtual-income term to equation (2), in the spirit of Gruber and Saez (2002). Two remarks follow for our design. First, households predicted to cross a bracket at the event also experience a larger increase in tax liability than their within-cell controls, so, absent a virtual-income instrument, our estimates should be read as total (uncompensated) responses to the event-induced tax change; an extension instrumenting a simulated virtual-income term would separate substitution from income responses. Second, the component of the needs change $\kappa_{it}(1) - \kappa_{it}(0)$ that is common within a comparison cell affects treated and control households alike and differences out; only a component covarying with threshold placement would threaten identification, which is the same requirement as for the direct effect on potential income.²⁴

The participation margin

The intensive-margin model cannot generate extensive-margin responses. Following H. J. Kleven et al. (2009), consider a spouse deciding whether to work at all, facing a fixed cost of participation q_{it} (commuting, childcare, forgone home production). Conditional on working,

²⁴Appendix G takes income effects seriously for a different purpose: they are one reason why first-stage monotonicity can fail in reform-based simulated-IV designs, though not in ours.

earnings are chosen optimally, so the value of participation is

$$V_{it}^w = \max_z \left\{ z - \left[T_t^H(z + z_{pt}; Q_{it}) - T_t^H(z_{pt}; Q_{it}) \right] - \psi(z; n_{it}) \right\},$$

where the bracketed term is the *participation tax*: the increase in the household's tax liability caused by the spouse's earnings, given partner income and the fiscal configuration. The spouse works if and only if

$$V_{it}^w \geq q_{it}.$$

The relevant price on this margin is therefore not the marginal rate at current earnings but the participation tax wedge—and its first-euro marginal rate, $1 - \tau^H$ evaluated at $z = 0$, is exactly what partner income and the quotient determine: because partner income alone places the household above the tax-free threshold, the second earner's first euro is taxed at the household's current bracket. A household pushed into a higher bracket by the child's fiscal exit thus sees the return to the second earner's participation fall at every level of own earnings, which rationalises the extensive-margin responses we document. With heterogeneous fixed costs q_{it} , the extensive-margin analogue of the elasticity is the response of the participation probability to the participation net-of-tax rate.

Appendix B Complementary material - case study

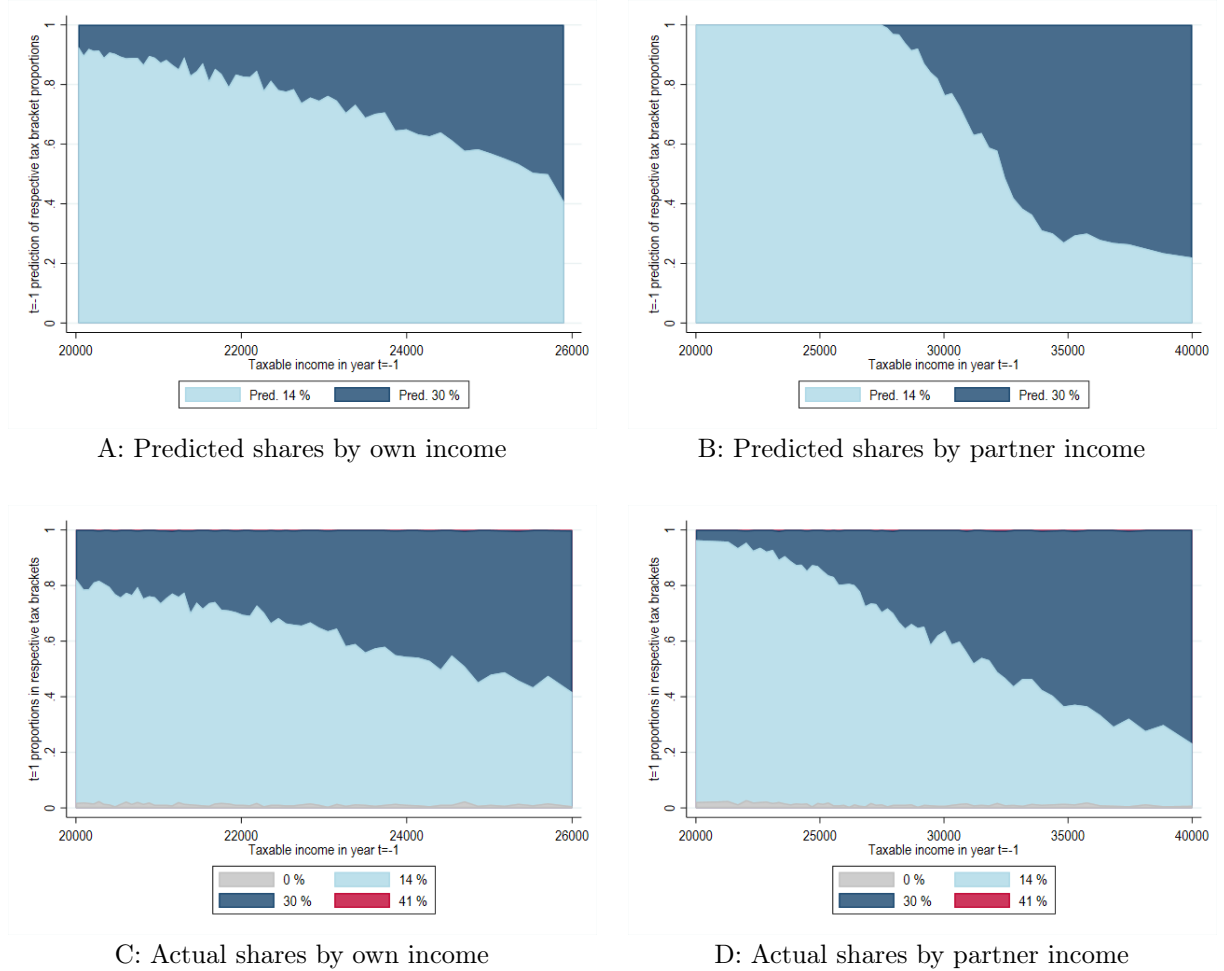


Figure B.1: Description of actual and predicted variation

Notes: Panel A describes the proportions of simulated treated (predicted to lie in the 30% tax bracket after marriage) and simulated control (predicted to lie in the 14% tax bracket) by own taxable income in year $t = 1$. The prediction is based on taxable income of both partners in $t = -1$. Panel B shows the corresponding proportions by partner taxable income. Panels C and D report the actual tax brackets in year $t = 1$, by own taxable income (Panel C) and by partner taxable income (Panel D).

Figure B.1 displays the predicted changes in marginal tax rates around the marriage event. For example, approximately 50% of individuals earning €24,000 in the year before marriage are predicted to experience a change in their marginal tax rate due to marriage, while the remaining 50% are not. There are two main reasons why the share of individuals experiencing an increase in the marginal tax rate rises with pre-marriage income. First, the distance to the kink decreases: if partner incomes were randomly distributed, this would mechanically

generate the observed pattern. Second, due to assortative matching, individuals with higher pre-marriage income are more likely to marry partners who also have higher income, thereby increasing the likelihood of crossing the kink and experiencing a change in the marginal tax rate due to marriage.

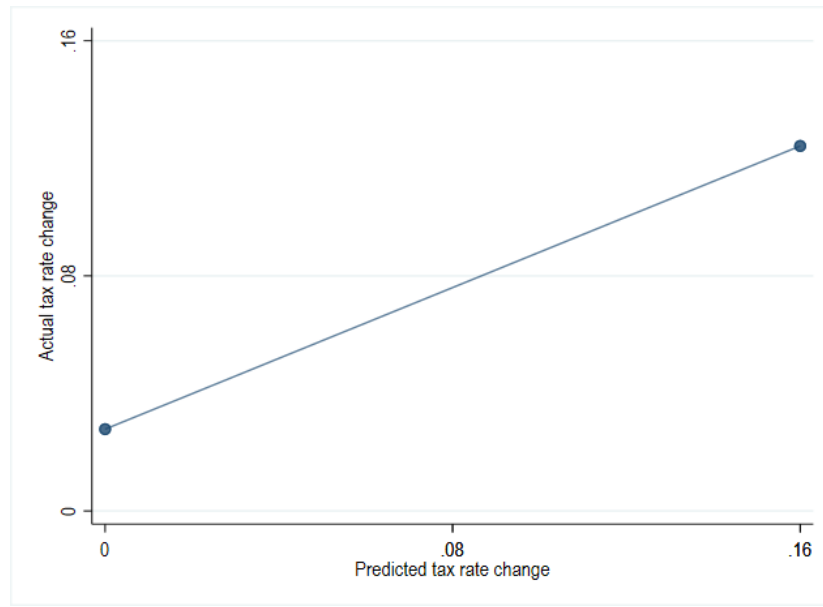


Figure B.2: First-stage relationship between predicted and actual tax changes

Notes: The figure shows the first-stage relationship between the simulated change in marginal tax rates induced by marriage and the realised change in marginal tax rates.

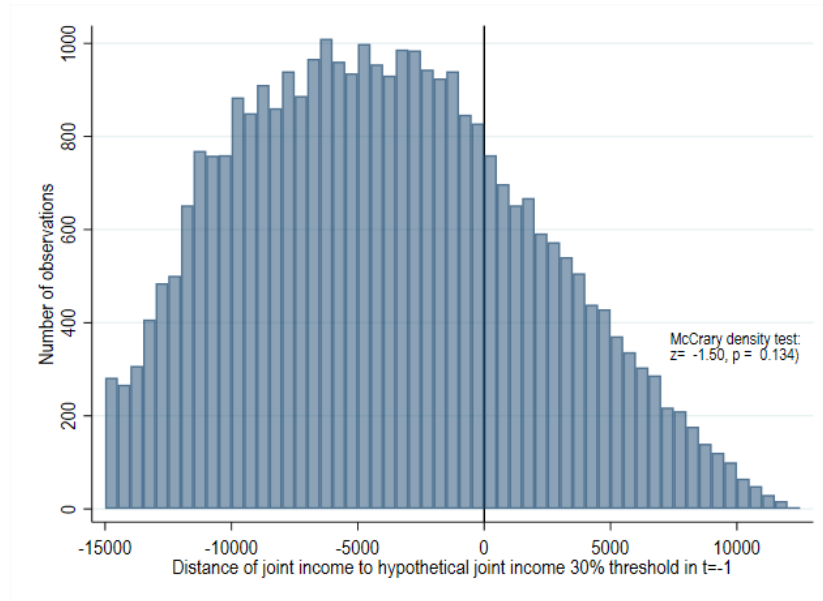


Figure B.3: McCrary density test at the hypothetical joint-income threshold

Notes: The figure plots the number of observations by distance to the joint-income threshold at which the marginal tax rate increases from 14% to 30%, for the case-study sample. It also reports the McCrary density test for excess mass at the threshold.

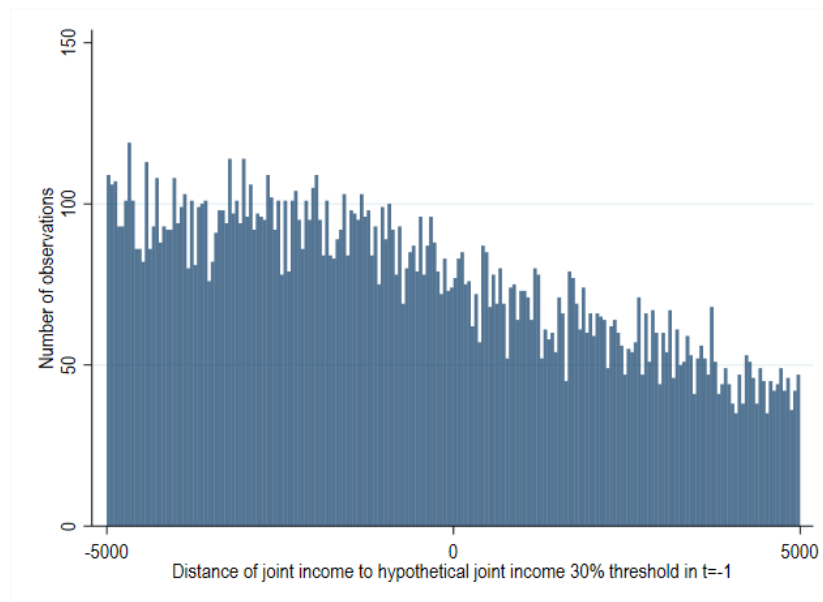


Figure B.4: Density of the running variable underlying the instrument

Notes: The figure displays the density of the running variable underlying the indicator instrument: distance to the hypothetical joint-income threshold.

This appendix collects the validity checks for the case-study cell of Section 4: the first stage relating predicted to realised bracket changes, and density tests for manipulation of the running variable.

First stage. Figure B.5 relates the predicted bracket change to the realised change in the household’s marginal rate around the exit. The prediction is strong but compliance is imperfect: because the prediction freezes incomes at their pre-event values, income growth and behavioural responses move some households across thresholds in both directions. This wedge between predicted and realised changes is why our estimands are defined on the assignment and rescaled explicitly (Section 4.3), with Wald versions rescaling by the realised first stage.

Figure B.5: Predicted and realised changes in the marginal tax rate around the child’s fiscal exit

Notes:

No anticipation. If households anticipated the exit and adjusted incomes in advance—bunching below the future $Q = 2$ threshold, or timing the detachment strategically—the pre-event incomes from which we compute the prediction would themselves be selected. Figure B.6 examines the density of the running variable underlying the binary prediction, household income per $Q = 2$ part in the pre-event year, around the relevant threshold. The McCrary test does not reject continuity, and there is no visible excess mass. Section 4 states the corresponding no-anticipation assumption.

Figure B.6: Density of pre-event household income around the $Q = 2$ bracket threshold

Notes:

Appendix C Complementary material - results

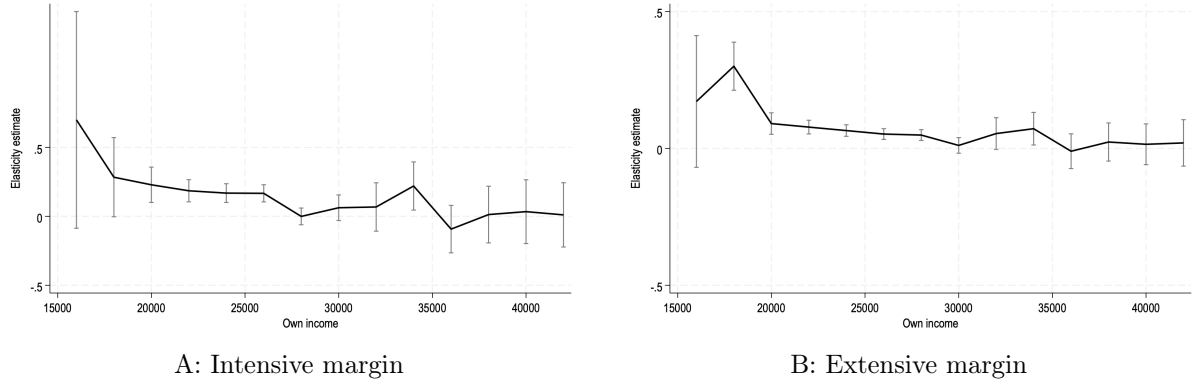


Figure C.1: Elasticities along the income distribution (OLS, women, positive tax changes)

Notes: The figure reports estimated elasticities of labour income along the pre-marriage income distribution for women facing positive simulated tax changes. Panel A shows intensive-margin elasticities, while Panel B shows extensive-margin elasticities, where the extensive margin is defined as an indicator for positive labour income.

Estimates are obtained from the OLS specification in Equation 10, estimated separately for each simulated tax change δ on the sample with $\Delta Z_i^{-1} \in \{0, \delta\}$. In contrast to the baseline matching specification, no restriction based on covariate proximity is imposed, and all observations in the relevant δ -sample are used.

Appendix D Robustness: the marriage design

Appendix E Extensive-margin estimates

Appendix F Event-time profiles

Appendix G Monotonicity

This appendix does three things. First, it states the monotonicity condition that is relevant when the instrument is a *constructed* variable, as in the simulated-IV literature and in our design: monotonicity of the realised treatment in the *primitive shock*, which is the condition under which the estimand is a convex—and hence weakly causal—average of individual causal effects (Angrist et al., 2000; Blandhol et al., 2022; Borusyak et al., 2022). Second, it shows by means of a calibrated example that this condition can fail in the canonical reform-based simulated-IV setting; this setting is a foil, not our identifying variation. Third, it argues that the condition is very likely to hold for our design, because the shock we exploit—the fiscal exit of a child—shifts the tax schedule outward without changing its gradient.

Although standard, the monotonicity assumption has, to our knowledge, received no explicit attention in the simulated IV literature. We characterise the conditions for a violation under the canonical setting in which simulated, reform-induced marginal tax rate changes affecting part of the schedule are used as instrumental variation (Gruber & Saez, 2002; Jakobsen & Søggaard, 2022).

In that setting, the primitive shock is the reform itself. Let $g \in \{0, 1\}$ indicate whether the reform is in force, and let $D_i(g)$ denote the marginal tax rate individual i faces under shock status g , inclusive of behavioural responses: the realised treatment is an equilibrium object, determined jointly by the schedule and by the individual’s choice of income under that schedule. Equivalently, $D_i(g) = D(g, \eta_i)$: the treatment is a structural function of the shock and an individual heterogeneity type η_i , as in Borusyak et al. (2022, appendix A.1), whose first-stage monotonicity condition $\partial x / \partial g \geq 0$ specialises, for a binary shock, to condition (G.1) below. The simulated instrument of the literature is the *mechanical prediction* of $D_i(1) - D_i(0)$ obtained by holding income fixed at its pre-reform ($t - k$) value; it is a deterministic function of

predetermined data. Because no individual responds to the researcher’s prediction, potential treatments indexed by the instrument are not well-defined causal objects, and the textbook Imbens–Angrist condition—that the instrument shift the treatment in the same direction for everyone (Imbens & Angrist, 1994)—cannot even be stated. The condition that matters instead operates one level down. The IV estimand can be written as a weighted average of individual causal responses, with weights proportional to the individual first-stage responses $\pi_i \equiv D_i(1) - D_i(0)$ (Angrist et al., 2000; Borusyak et al., 2022). If the π_i share a common sign, the weights are non-negative and the estimand is *weakly causal*: a convex average of causal effects that cannot reverse the sign of a uniformly signed effect (Blandhol et al., 2022). For a reform that raises statutory rates on part of the schedule, first-stage monotonicity therefore requires

$$(G.1) \quad D_i(1) \geq D_i(0) \quad \forall i,$$

with the reverse inequality for a reform that cuts rates. A violation—individuals whose realised marginal tax rate moves *against* the direction of the reform—puts negative weight on their responses, and the estimand loses its interpretation as a convex average.

Under the strong assumptions standard in the literature of (i) a quasi-linear utility model, implying no income effects, (ii) continuous labour supply, and (iii) no heterogeneity in elasticities, condition (G.1) is not violated. The intuition is that, for any reform—whether it increases or decreases the marginal tax rate in a given bracket—the utility ranking between the kink point and the interior solution remains unchanged. As a result, any new interior optimum in a bracket unaffected by the reform is dominated by the kink point at the reform-affected bracket.

When these assumptions are relaxed—by allowing for income effects, labour supply rigidities, and heterogeneity in elasticities or preferences—first-stage monotonicity can no longer be taken for granted.²⁵

²⁵Responses on the extensive margin are a potentially important source of violations. Extensive-margin responses to tax increases are common, as we also document in this paper, and they mechanically reduce the marginal tax rate, as individuals drop out of the labour market and thus move into a lower tax bracket.

A calibrated example

We illustrate a violation of first-stage monotonicity using a model with income effects and heterogeneity in the curvature of the utility function. Instead of the standard quasi-linear utility function, we consider a more general specification that allows for income effects and non-linear labour supply responses.²⁶ Individuals derive utility from consumption c and hours worked h according to

$$u_i(c, h) = \frac{c^{1-\sigma_i}}{1-\sigma_i} - \chi_i \frac{h^{1+\varphi_i}}{1+\varphi_i},$$

where $\sigma_i > 0$ governs the curvature of utility from consumption, and hence the strength of income effects, $\varphi_i > 0$ governs the curvature of the disutility from labour, and $\chi_i > 0$ scales the marginal disutility of hours worked. Labour income is given by $y = wh$. We assume that hours rigidities restrict individuals to four discrete choices: 0, 20, 30, or 40 hours per week.

For the tax schedule, we consider the French income tax system as of 2015. In the part of the schedule on which we focus, labour income up to €26,791 per year is taxed at a marginal rate of 14 %, and income above this threshold at 30 %.²⁷ Because utility is defined over hours worked, we can equivalently express the tax schedule as a function of hours, with a kink at $h = k$ corresponding to the income threshold. In the example, we set the hourly wage to €20.6, so that the threshold at €26,791 corresponds to approximately 25 hours of work per week.

We study a hypothetical reform ($g = 1$) that eliminates the 30 % bracket by raising the marginal tax rate above the threshold from 30 % to 41 %, equal to the next statutory rate in the French schedule.

We consider two preference types, Type A and Type B, and calibrate the model such that both share the same discrete optimum of 30 hours per week under no reform. The key parameter governing the relative strength of income and substitution effects is the consumption-curvature parameter σ_i . Type A is calibrated with $\sigma_A < 1$, so that the substitution effect dominates, whereas Type B is calibrated with $\sigma_B \gg 1$, so that the income effect dominates. For each type, the disutility scale χ_i is chosen to match the discrete optimum of 30 hours per week

²⁶Hours rigidities would be sufficient to illustrate a violation. To make the argument applicable more broadly, however, we consider a more general model with income effects.

²⁷To simplify the exposition, we ignore the *décote*, which does not affect the example, as can be shown upon request.

under no reform. The resulting preference parameters are

$$(\sigma_A, \varphi_A, \chi_A) = (0.696, 0.300, 0.272), \quad (\sigma_B, \varphi_B, \chi_B) = (7.000, 0.300, 9.17 \times 10^{-30}).$$

Under no reform, both types choose 30 hours per week, placing them in the 30 % bracket:

$$D_A(0) = D_B(0) = 0.30.$$

The simulated instrument, computed by holding both types' incomes fixed at their 30-hour values, accordingly predicts the same treatment change for both: a rise in the marginal tax rate from 30 % to 41 %, that is, a predicted first-stage response of +0.11.

The optimal hours choices under the reform and under no reform are shown in Figure G.1.²⁸

Under the reform, the two types' optimal choices diverge:

$$h_A^* = 20 \text{ hours/week}, \quad h_B^* = 40 \text{ hours/week}.$$

Type A chooses 20 hours under the reform, moving below the bracket threshold and facing the lower marginal tax rate of 14 %. Type B chooses 40 hours, moving above the threshold and facing the reform rate of 41 %.

The realised treatments therefore respond to the same shock in opposite directions:

$$\pi_A = D_A(1) - D_A(0) = 0.14 - 0.30 = -0.16 < 0, \quad \pi_B = D_B(1) - D_B(0) = 0.41 - 0.30 = +0.11 > 0.$$

Type A's marginal tax rate moves against the direction of the reform: despite being exposed to a shock that raises statutory rates—and despite the instrument predicting a tax increase for it—its realised marginal tax rate *falls*, violating condition (G.1). Type B's realised treatment moves with the reform.

First-stage monotonicity is therefore violated. The IV estimand places negative weight on the responses of individuals like Type A, is no longer a convex average of causal effects, and is

²⁸The optimal hours choice under each regime is determined by maximising $u_i(c, h)$ subject to the piecewise-linear budget constraint. There is no closed-form solution in general, so the optimum is obtained numerically by comparing utility across all candidate solutions: interior first-order-condition roots on each linear segment, the kink, and the corner at zero hours.

not weakly causal in the sense of Blandhol et al. (2022): with heterogeneous elasticities it can even reverse the sign of a uniformly signed behavioural response.

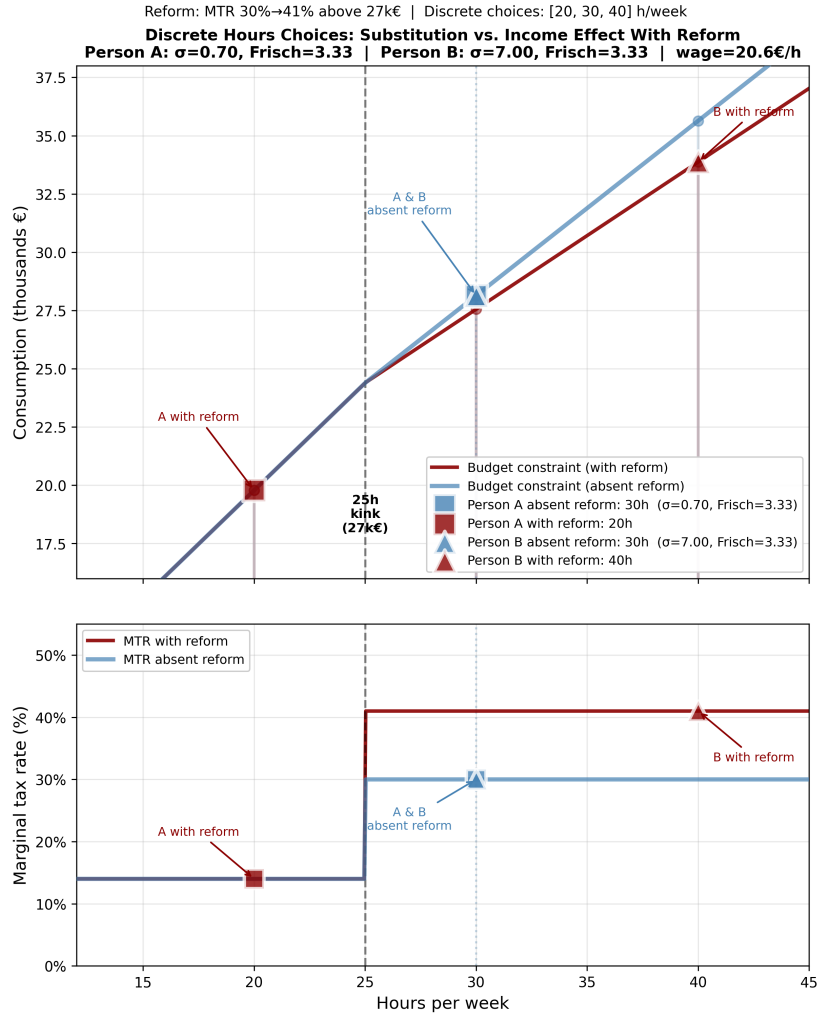


Figure G.1: Violation of first-stage monotonicity under heterogeneous labour-supply responses

Top panel. Budget constraints under the reform (red) and under no reform (blue). The reform raises the marginal tax rate above the second bracket threshold from 30 % to 41 %. Both Type A and Type B optimally choose 30 h/week under no reform. Under the reform, Type A chooses 20 h/week and is below the kink; Type B ($\sigma_B = 7$) chooses 40 h/week and is above the kink.

Bottom panel. Marginal tax rates under the reform (red) and under no reform (blue), as a function of hours worked, with the same optimal choices indicated.

It is worth noting that a strong income effect for Type B is not necessary for the violation. Even in the absence of income effects, Type B could simply choose 30 hours per week under the reform as well and be in the now higher 41 % bracket. The same violation would then

arise: Type A chooses 20 hours under the reform and is in the 14 % bracket ($\pi_A = -0.16$), while Type B stays at 30 hours ($\pi_B = +0.11$).

Models with extensive-margin labour supply responses can also generate violations of first-stage monotonicity. An extensive-margin response to, for example, a reform that increases tax rates moves an individual out of the tax schedule altogether, that is, in France, to a marginal tax rate of zero, so that $\pi_i < 0$. If, at the same time, other individuals remain employed and face the higher rate under the reform, $\pi_i > 0$, the realised treatment responds to the shock in opposite directions across individuals, violating condition (G.1). Because extensive-margin responses to tax increases are empirically common, as we also document in this paper, this is a realistic and quantitatively important source of monotonicity violations in reform-based IV strategies.

We stress that the reform-based setting of this example is a foil: it is the canonical design from which our identification strategy departs, not our identifying variation. We now turn to our design.

Why first-stage monotonicity is likely satisfied for our instrument

In our design, the primitive shock is the fiscal exit of the first-born child, which reduces the household's fiscal parts from $Q = 2.5$ to $Q = 2$. Writing $g_{ir} = \mathbb{1}\{r \geq 0\}$ for whether the exit has occurred by event year r , let $D_{ir}(g)$ denote the marginal tax rate the household would face at event time r with ($g = 1$) and without ($g = 0$) the exit, inclusive of behavioural responses: the potential-treatment concept indexed by the primitive shock, never by the constructed prediction ΔZ_i .

Two remarks on the epistemic status of this object. First, only one arm is ever observed: $D_{ir}(1)$ is the realised bracket for $r \geq 0$, while $D_{ir}(0)$ —the bracket the household would face had the child not exited—is counterfactual for every household in our sample, since all experience the exit. The simulated instrument ΔZ_i is a mechanical *prediction* of the contrast $D_{i0}(1) - D_{i0}(0)$, obtained by freezing incomes at their pre-event values and applying the two schedules; it is informative about, but not equal to, the behavioural contrast. Second, because $D_{ir}(0)$ is unobserved, the monotonicity condition below is not directly testable. In our design it is instead secured a priori: for the compliers in question, the unobserved arm equals the

(wide) pre-event bracket, $D_{ir}(0) = d_b$, and the floor argument bounds the observed arm from below by that same value, $D_{ir}(1) \geq d_b$.

A key feature of this shock is that it does not change the gradient of the tax schedule—as the reform above does—but shifts an otherwise unchanged schedule outward. Because household taxable income is divided by the number of parts, the marginal rate at *every* level of household income y satisfies

$$T'_t\left(\frac{y}{2}\right) \geq T'_t\left(\frac{y}{2.5}\right) :$$

the exit weakly raises the marginal tax rate at every point of the budget set. Two implications follow. First, the shock can only predict tax increases, $\Delta Z_i \geq 0$, so only the direction of condition (G.1) stated above is at stake. Second, a violation cannot arise mechanically from the shift itself; it would require a behavioural relocation, namely a household reducing its income by enough that its post-exit marginal rate falls strictly below its pre-exit rate—the Type A channel of the example above.

Partner income essentially rules this channel out. Figure G.2 illustrates for a given level of partner income. It shows the marginal tax rate an individual faces as a function of own income, before the exit ($Q = 2.5$, blue dashed) and after the exit ($Q = 2$, red solid), for a partner earning €40,000. The post-exit curve lies weakly above the pre-exit curve everywhere. Crucially, the marginal tax rate after the exit remains strictly positive—at 14 %—even when own income is zero, because the partner’s income alone, divided by the remaining two parts, already places the household above the tax-free threshold. An extensive-margin response, the most extreme possible reduction in own labour supply, therefore still leaves the household facing a 14 % marginal rate rather than a 0 % rate. Since our instrument captures individuals who are predicted to move from the 14 % to the 30 % bracket, a violation of first-stage monotonicity would require the realised marginal rate to fall strictly below 14 % upon the exit. This is impossible as long as partner income, net of deductions, exceeds twice the tax-free threshold per part (approximately €19,400 under the schedule of our example)—a condition we can verify cell by cell, since our cells fix the partner-income range.²⁹ Hence $D_{ir}(1) \geq D_{ir}(0)$ for all individuals in the cells we use: there are no individuals whose realised treatment moves

²⁹The residual channel is a simultaneous collapse of *both* spouses’ incomes below the household tax-free threshold. Deductions available upon the exit, such as support payments to the departed child, attenuate the first stage but respect the floor for the same reason: they are capped, and the floor argument applies to partner income net of the capped deduction.

against the shock, and by the argument of the previous section the estimand aggregates causal responses with non-negative weights.

Figure G.2: Marginal tax rate before and after the child’s fiscal exit, partner earns €40,000

Notes: The figure shows the effective marginal tax rate as a function of own gross income before the first-born’s fiscal exit ($Q = 2.5$, blue dashed) and after the exit ($Q = 2$, red solid), for a partner earning €40,000, based on the French income tax schedule without the *décote*. The post-exit schedule lies weakly above the pre-exit schedule at every income level, and equals 14 % even at zero own income because partner income alone places the household above the tax-free threshold. Even a complete exit from the labour market therefore cannot lower the marginal rate below 14 %, ruling out realised rate movements against the shock for households predicted to move from the 14 % to the 30 % bracket.

The same logic applies to the marriage-based robustness design, with one addition. Marriage to a higher-earning partner under joint taxation likewise shifts the tax liability function outward without changing its gradient, because partner income is added to the household tax base: partner income places a floor on the marginal tax rate an individual faces after marriage, as Figure G.3 illustrates. Unlike the child-exit design, however, marriage can predict tax *decreases* as well as increases. For predicted decreases, violations of first-stage monotonicity are ruled out symmetrically, since they would require a dramatic increase in labour supply.

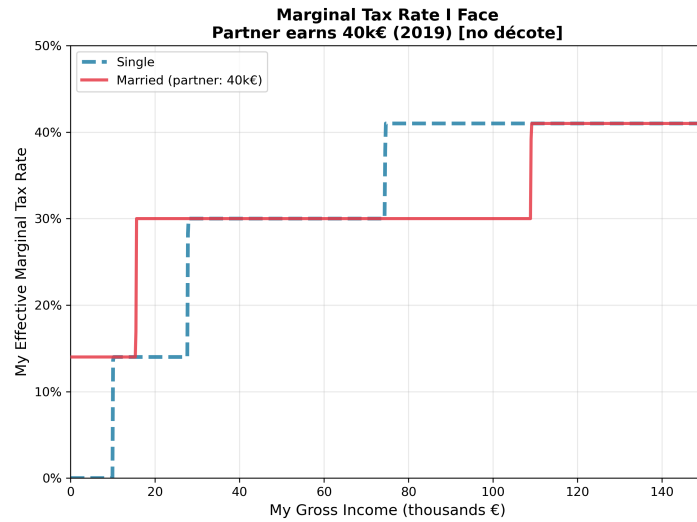


Figure G.3: Individual budget constraint and marginal tax rate under marriage, partner earns €40,000 (robustness design)

Notes: The figure shows the effective marginal tax rate as a function of own gross income under individual taxation (blue dashed) and under joint taxation when married to a partner earning €40,000 (red solid), based on the 2019 French income tax schedule without the *décote*. Income is not pooled, so own consumption corresponds to own after-tax income only. As a result, the marginal tax rate for a married individual is 14 % even at zero own income, because the partner's income of €40,000 already places the household above the tax-free threshold. Consequently, even a complete exit from the labour market cannot lower the individual's marginal tax rate below 14 %, ruling out a violation of first-stage monotonicity for individuals predicted to move from the 14 % to the 30 % bracket.